

CATEGORICAL LOCAL LANGLANDS AT THE STEINBERG PARAMETER: A CASE STUDY

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ABSTRACT. We study the categorical local Langlands correspondence for GL_n at the Steinberg parameter and compute the stalks of the corresponding Vogan sheaves at all basic points. Under the expected Eisenstein compatibility, this yields a complete formula for these stalks in terms of generalized Steinberg representations of the relevant inner forms. As a consequence, we obtain a general and effective formula for the compactly supported cohomology of admissible loci in Schubert cells of the B_{dR} -affine Grassmannian, hence for a large class of p -adic period domains. In particular, the cohomology is shown to split as a direct sum of shifted irreducible representations, and can be computed by an explicit algorithm. This gives, beyond the classical Drinfeld and Schneider–Stuhler cases, the first general cohomological formula for these spaces, marking an important milestone in a basic problem that has remained open for more than thirty years. We also compute the Ext-groups between Vogan sheaves and use them to construct canonical extensions with striking structural properties.

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1. INTRODUCTION

In this paper, we make a detailed study of the categorical local Langlands correspondence for GL_n around the so-called Steinberg parameter. Recall that categorical local Langlands conjecture (CLLC) predicts an equivalence of categories

$$\mathbf{L}_\psi: \mathrm{D}(\mathrm{Bun}_G) \xrightarrow{\sim} \mathrm{IndCoh}(\mathrm{Par}_G)$$

for any quasisplit group G , depending only on a choice of Whittaker datum ψ and satisfying many good properties.

Given a semisimple parameter φ , it is possible to localize the equivalence $\mathbf{L}_\psi$ around φ . The complexity of this localized equivalence is controlled by the geometry of the *Vogan stack* $V_\varphi = \tilde{V}_\varphi/S_\varphi \subseteq \mathrm{Par}_G$. In particular, there are two notable cases where the complexity of CLLC is manageable:

- (i) When $\tilde{V}_\varphi$ is a point and S_φ° is a torus, we are exactly in the situation where φ is *generous*. In this case, we expect that the semiorthogonal decomposition on $\mathrm{D}(\mathrm{Bun}_G)$ splits canonically, and that Hecke operators are perverse t-exact. Moreover, we expect an explicit matching between some very simple sheaves on both sides encoding the classical local Langlands correspondence.
- (ii) When $\tilde{V}_\varphi$ is a point, we are exactly in the situation where φ is generic (in the technical sense). In this case, we still expect that the Hecke action is perverse t-exact. However, in general the semiorthogonal decomposition does not split, and there are several canonical families of sheaves on both sides whose matching should encode classical LLC.

The previous two cases were studied carefully in [Han23], where the second author formulated a number of conjectures describing CLLC localized around such parameters. In this paper, we turn to the complementary situation where S_φ° is a torus but $\tilde{V}_\varphi$ is arbitrary. Here the story is much more complicated, and our goal here is to make a detailed analysis in one paradigmatic case: we will assume that $G = \mathrm{GL}_n$ and that φ is the *Steinberg parameter*, i.e. the unramified parameter $\varphi(w) = \mathrm{diag}(|w|^{(n-1)/2}, |w|^{(n-3)/2}, \dots, |w|^{(1-n)/2})$. For this parameter, $\tilde{V}_\varphi = \mathbb{A}^{n-1}$ and $S_\varphi = \hat{T}$ is the usual diagonal maximal torus. This parameter is especially interesting for several reasons:

- ◊ In a precise sense, it is the most singular point of $\mathrm{Par}_{\mathrm{GL}_n}$.

- ◊ The Fargues–Scholze packet over φ contains the Steinberg representation, which (up to twist) is the only unipotent discrete series representation of GL_n . It also contains the trivial representation.

When $n = 2$, all the results we describe were obtained by Bertoloni Meli–Koshikawa several years ago, and their work has been a great inspiration and guide to the second author. However, when $n > 2$, we shall see some substantial new phenomena.

1.1. Stalks of Vogan sheaves: conjectures. We now turn to a more precise description of the conjectures which motivated this paper. Fix $G = \mathrm{GL}_n$ and let φ be the (semisimplified) Steinberg parameter as above. For any $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{X}^*(\hat{T}) = \mathbb{Z}^n$, we have the obvious weight λ line bundle $\mathcal{O}(\lambda)$ on $V_\varphi = \mathbb{A}^{n-1}/\hat{T}$. Letting $\iota: V_\varphi \hookrightarrow \mathrm{Par}_G$ be the canonical closed immersion, we set $\mathcal{L}_\lambda = \iota_* \mathcal{O}(\lambda) \in \mathrm{Coh}(\mathrm{Par}_G)$. These sheaves, which we call *Vogan sheaves*, are the star players in this paper. Under CLLC, they correspond to a family of automorphic sheaves $\mathcal{L}_\lambda^{\mathrm{aut}} \in \mathrm{D}(\mathrm{Bun}_G)$. Alternatively, in Zhu’s formulation of CLLC they correspond to a family $\mathcal{L}_\lambda^{\mathrm{Isoc}} \in \mathrm{D}(\mathrm{Isoc}_G)$.

Our first goal is a description of the stalk of $\mathcal{L}_\lambda^{\mathrm{aut}}$ at the point $b = 1$. To explain this, we need a little more notation. For any $I \subseteq \{1, 2, \dots, n-1\}$, let $N_I = \sum_{i \in I} E_{i, i+1}$ where E_{ij} is the usual elementary matrix. As I varies, N_I enumerates the orbits in V_φ . Moreover, (φ, N_I) is a Frobenius-semisimple Weil–Deligne parameter, so it has a unique matching irreducible smooth representation π_I under the classical local Langlands correspondence. It is not hard to show that the representations π_I are exactly the irreducible quotients of the smooth representations $\mathcal{C}(P(E) \backslash G(E))$ as P varies over all parabolics in G . Following standard terminology, we call the π_I *generalized Steinberg representations*. In particular, $\pi_\emptyset = \mathbf{1}$ is the trivial representation and $\pi_{\{1, 2, \dots, n-1\}} = \mathrm{St}$ is the Steinberg representation.

Let $d = \sum_i \lambda_i$ be the degree of λ . It is easy to see that $i_1^* \mathcal{L}_\lambda^{\mathrm{aut}}$ vanishes identically unless λ has degree zero. When λ does have degree zero, we can uniquely write $\lambda = \sum_{1 \leq i \leq n-1} m_i \alpha_i$ for some integers m_i , where α_i are the usual simple roots. In this case we have the following conjecture.

Conjecture 1.1.1. *For any degree zero weight $\lambda = \sum_{1 \leq i \leq n-1} m_i \alpha_i$, we have*

$$i_1^* \mathcal{L}_\lambda^{\mathrm{aut}} = \pi_I[\sum_{i \in I^c} (1 - 2m_i)]$$

where $I = \{1 \leq i \leq n-1 : m_i \leq 0\}$.

Some cases of this were known previously. In particular, Hellmann showed that $\mathcal{L}_0$ should correspond to $i_1^* \mathrm{St}$, and in fact he fully computed $\mathcal{L}_\lambda^{\mathrm{aut}}$ for any degree zero weight $\lambda = \sum m_i \alpha_i$ with all $m_i \in \{0, 1\}$. However, it came as a great surprise to the second author that $i_1^* \mathcal{L}_\lambda^{\mathrm{aut}}$ is *always* a shifted irreducible representation.

In fact, the same phenomenon persists at any basic point of Bun_G . To explain this, pick any integer d , and let $b_{d/n} = b_d$ be the basic isocrystal of rank n and slope d/n . Write $n' = n/(d, n)$ and $d' = d/(d, n)$, where we declare $(0, n) = n$. Then $G_{b_d} = \mathrm{GL}_{(d, n)}(D_{d'/n'})$ is an inner form of G (which depends only on $d \bmod n$). This inner form comes with its own set of generalized Steinberg representations ρ_I which are in canonical bijection with subsets $I \subseteq \{1, 2, \dots, (d, n) - 1\}$.¹ Explicitly, under local Langlands ρ_I corresponds to the L-parameter (φ, N) where N is the

¹We call any such I *d-relevant* to emphasize its dependence on d .

monodromy supported on the superdiagonal with $N_{i,i+1} = 1$ except when $i = n'j$ with $j \notin I$, in which case $N_{i,i+1} = 0$. Again, when $I = \{1, 2, \dots, (d, n) - 1\}$ is maximal, $\rho_I = \text{St}_{G_{b_d}}$ is the Steinberg representation. By the results of [HKW22], the ρ_I are exactly the irreducible $G_{b_d}(E)$ -representations with semisimple L-parameter φ .

It will be very important to single out the following basepoint weight. For a given d , let $\omega_d \in \mathbb{X}^*(\hat{T})$ be the *Steinberg weight vector* of G_{b_d} , defined explicitly as

$$\omega_d := \left(\left\lceil \frac{d}{n} \right\rceil, \left\lceil \frac{2d}{n} \right\rceil - \left\lceil \frac{d}{n} \right\rceil, \left\lceil \frac{3d}{n} \right\rceil - \left\lceil \frac{2d}{n} \right\rceil, \dots, d - \left\lceil \frac{(n-1)d}{n} \right\rceil \right).$$

Note that $\omega_d = 0$ if $d = 0$, and ω_d is central exactly when $n \mid d$. If $d = 1$ then $\omega_d = (1, 0, \dots, 0)$. If $n = 5$ then $\omega_2 = (1, 0, 1, 0, 0)$ and $\omega_3 = (1, 1, 0, 1, 0)$. Note that ω_d has degree d , so if λ is any weight of degree d , we can uniquely write $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ for some integers m_i . We then have the following generalization of Conjecture 1.1.1.

Conjecture 1.1.2. *If $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ is any weight of degree d , we have*

$$i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}} = \rho_I[\sum_{1 \leq i \leq n-1, m_i > 0} (\delta_i - 2m_i)].$$

Here $I = \{0 < j < (d, n) : m_{n'j} \leq 0\}$ and $\delta_i \in \{0, 1\}$ with $\delta_i = 1$ if and only if $n' \mid i$.

Example 1.1.3. When $(d, n) = 1$, the inner form G_{b_d} is anisotropic mod center, the only d -relevant I is the empty set, and the only ρ_I is the trivial representation $\mathbf{1}$. In this case the conjecture specializes to

$$i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}} = \mathbf{1}[-\sum_{1 \leq i \leq n-1, m_i > 0} 2m_i].$$

with $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ as above.

A degree d weight $\omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ is *basic* if $m_i \in \{0, 1\}$ and $m_i = 0$ for all $n' \nmid i$. In fact, the degree d basic weights are in canonical bijection with d -relevant I 's, by sending $I \subseteq \{1, \dots, (d, n) - 1\}$ to $\omega_d + \sum_{j \in I^c} \alpha_{n'j}$. In particular, $I = \{1, \dots, (d, n) - 1\}$ goes to ω_d , and $I = \emptyset$ goes to $\omega_d + \sum_{0 < j < (d, n)} \alpha_{n'j}$. At these weights, and *only* these weights, we expect that $\mathcal{L}_\lambda^{\text{aut}}$ is $!$ -extended from the point b_d . Assuming CLLC, this has been proved by Bertoloni Meli–Koshikawa. In this case, Conjecture 1.1.2 specializes to the following.

Conjecture 1.1.4. *If $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ is a basic weight of degree d , with I the associated d -relevant set, then*

$$\mathcal{L}_\lambda^{\text{aut}} = i_{b_d}^* \rho_I[-|I^c|].$$

Example 1.1.5. When I is maximal among the d -relevant sets, ρ_I is the Steinberg representation of G_{b_d} . So Conjecture 1.1.4 specializes to the prediction that

$$\mathcal{L}_{\omega_d}^{\text{aut}} = i_{b_d}^* \text{St}_{G_{b_d}},$$

explaining the name for Steinberg weight ω_d .

When $d = 0$, we have $(d, n) = n$ and $G_{b_d} = G$, and the prescribed bijection sends $I \subseteq \{1, \dots, n-1\}$ to the basic weight $\alpha_{I^c} := \sum_{i \in I^c} \alpha_i$. Then Conjecture 1.1.4 specializes to the prediction that

$$\mathcal{L}_{\alpha_{I^c}}^{\text{aut}} = i_{1!} \pi_I[-|I^c|].$$

This matching is already proved by Hellmann, see [Hel23, Remark 4.43].

1.2. Stalks of Vogan sheaves: results. Our first main result is the following theorem.

Theorem 1.2.1. *Assume the Eisenstein compatibility, i.e., the functor*

$$c_\psi: D(\text{Bun}_G) \longrightarrow \text{QCoh}(\text{Par}_G)$$

is compatible with Eisenstein series as in [HM26, Conjecture 1.4.2]. Then Conjecture 1.1.2 is true for all weights.

Note that under the assumption stated here, the work [HM26] proves the full CLLC for GL_n .

For Isoc_G , we also unconditionally prove the analogue of Conjecture 1.1.1.

Theorem 1.2.2. *For any degree zero weight $\lambda = \sum m_i \alpha_i$, we have*

$$i_1^! \mathcal{L}_\lambda^{\text{Isoc}} = \pi_I[\sum_{i \in I^c} (1 - 2m_i)]$$

where $I = \{1 \leq i \leq n - 1 : m_i \leq 0\}$.

We first sketch the proof of Theorem 1.2.1 for degree zero weights. Here the idea is very simple. For soft reasons, we know that $i_1^* \mathcal{L}_\lambda^{\text{aut}}$ is built entirely from the representations π_I . Now we observe that the π_I are neatly separated by their Jacquet modules. More precisely, it is not difficult to prove that $r_G^B(\pi_I) = \bigoplus_{\sigma: \text{Asc}(\sigma)=I} \sigma \delta^{1/2}$. Here $\text{Asc}(\sigma) = \{i \in [n - 1] : \sigma(i) < \sigma(i + 1)\}$ denotes the ascent set of the permutation σ . Note that the constituents of these Jacquet modules are pairwise-disjoint as I varies. This implies that to prove Theorem 1.2.1 in degree zero, it's enough to compute $r_G^B i_1^* \mathcal{L}_\lambda^{\text{aut}}$ and see that it matches the Jacquet module of $\pi_I[\sum_{i \in I^c} (1 - 2m_i)]$.

Next, we note that

$$r_G^B i_1^* \mathcal{L}_\lambda^{\text{aut}} = i_1^{T*} \text{CT}_{B!} \mathcal{L}_\lambda^{\text{aut}}$$

by the results in [HHS24]. Now we move this to the spectral side. Our assumption of Eisenstein compatibility in CLLC implies a similar constant term compatibility, so we can now translate this to the problem of analyzing $\text{CT}_B^{\text{spec}} \mathcal{L}_\lambda \in \text{IndCoh}(\text{Par}_T)$. For formal reasons, this sheaf is set-theoretically supported at the orbit $\{\sigma\varphi\} \subseteq |\text{Par}_T|$, so it inherits a canonical direct sum decomposition $(\text{CT}_B^{\text{spec}} \mathcal{L}_\lambda)^\sigma$ by supports. It is then easy to see the following key thing:

- (†) For any $i \in \mathbb{Z}$ and $\sigma \in S_n$, the multiplicity of $\sigma \delta^{1/2}$ in $H^i(r_G^B i_1^* \mathcal{L}_\lambda^{\text{aut}})$ equals the dimension of $H^i(R\Gamma(\text{Par}_T, (\text{CT}_B^{\text{spec}} \mathcal{L}_\lambda)^\sigma))$.

This is still not very explicit. The next step is to localize on Par_B . For this, let X be the formal completion of Par_G along V_φ , and let $\tilde{X}$ be the formal completion of Par_B along the preimage of V_φ , which is equivalently the fiber product $\text{Par}_B \times_{\text{Par}_G} X$. Note that $\tilde{X}$ clearly maps to the formal completion of Par_T along the orbit $\{\sigma\varphi\} \subseteq |\text{Par}_T|$, and it turns out that the connected components of $\tilde{X}$ are exactly the fibers of the individual points in this orbit. We write $\tilde{X}_\sigma$ for the connected component living over $\sigma\varphi$. The natural map $\text{Par}_B \rightarrow \text{Par}_G$ induces a proper map $p: \tilde{X} \rightarrow X$, and the composite map $p_\sigma: \tilde{X}_\sigma \rightarrow \tilde{X} \xrightarrow{p} X$ turns out to be a closed immersion for all σ . Some formal manipulations then give the following refinement of the observation (†) above.

- (‡) For any $i \in \mathbb{Z}$ and $\sigma \in S_n$, the multiplicity of $\sigma \delta^{1/2}$ in $H^i(r_G^B i_1^* \mathcal{L}_\lambda^{\text{aut}})$ equals the dimension of $H^i(R\Gamma(\tilde{X}_\sigma, p_\sigma^! \iota_* \mathcal{O}(\lambda))) = H^i(R\text{Hom}(p_{\sigma*} \mathcal{O}_{\tilde{X}_\sigma}, \iota_* \mathcal{O}(\lambda)))$.

Here we slightly abusively write $\iota: V_\varphi \rightarrow X$ for the factorization of ι through the formal completion. This reduces us to computing the complex $R\mathrm{Hom}(p_{\sigma*}\mathcal{O}_{\tilde{X}_\sigma}, \iota_*\mathcal{O}(\lambda))$. Note that the diagram $\tilde{X}_\sigma \rightarrow X \leftarrow V_\varphi$ is not tor-independent, and neither of the closed immersions here is a regular immersion. On the other hand, all the stacks here can be written down very explicitly, following ideas of Hellmann [Hel23] and X. Yang [Yan25]: there is an isomorphism

$$X \simeq (\mathrm{Spf} k[[u_1, \dots, u_n]][v_1, \dots, v_{n-1}]/(u_1v_1, \dots, u_{n-1}v_{n-1}))/\hat{T}$$

where $\hat{T}$ acts on v_i through the weight α_i , under which the substacks $\tilde{X}_\sigma$ resp. V_φ are cut out by setting $v_i = 0$ for all $i \notin \mathrm{Asc}(\sigma)$ resp. $u_i = 0$ for all i . At this point, setting

$$R = k[[u_1, \dots, u_n]][v_1, \dots, v_{n-1}]/(u_1v_1, \dots, u_{n-1}v_{n-1}),$$

we roll up our sleeves and write down an explicit $\hat{T}$ -equivariant projective resolution $\mathcal{P}_\bullet^{I^c}$ of $R/(v_i)_{i \in I^c}$ by finite free R -modules, for any fixed I . This resolution is necessarily unbounded, and the rank of C_j^I grows asymptotically like $cj^{|I^c|}$ for large j . As such, it takes some care to even guess this resolution, let alone to write it down in a manageable form and use it in calculations. However, in the end, explicit computations with this resolution give exactly what we need:

Proposition 1.2.3. *If $\lambda = \sum m_i \alpha_i$ is a degree zero weight and $I = \{i \text{ with } m_i \leq 0\}$, then we have $R\mathrm{Hom}(p_{\sigma*}\mathcal{O}_{\tilde{X}_\sigma}, \iota_*\mathcal{O}(\lambda)) = 0$ for all σ such that $\mathrm{Asc}(\sigma) \neq I$, while for all σ with $\mathrm{Asc}(\sigma) = I$ we have*

$$R\mathrm{Hom}(p_{\sigma*}\mathcal{O}_{\tilde{X}_\sigma}, \iota_*\mathcal{O}(\lambda)) = k[\sum_{i \in I^c} (1 - 2m_i)].$$

We note that an exact analogue of $(\dagger)$ holds in the Isoc setting, so the same calculation implies Theorem 1.2.2 with no additional work.

1.3. Cohomology of period domains. Our results have decisive applications towards computing the cohomology of p -adic period domains. To explain this, let μ be any cocharacter for G , and let $\mathrm{Gr}_{G, \leq \mu}$ be the associated Schubert cell in the B_{dR} -affine Grassmannian. Any point in this Schubert cell induces a μ -bounded modification $\mathcal{E}_x$ of the trivial rank n vector bundle on the Fargues–Fontaine curve, and we write $\mathrm{Gr}_{G, \leq \mu}^{\mathrm{adm}}$ for the $G(E)$ -stable open subspace where $\mathcal{E}_x$ is semistable. In the most classical example where $\mu = (1, 0, \dots, 0)$, we have $\mathrm{Gr}_{G, \leq \mu} = \mathbb{P}^{n-1}$, and $\mathrm{Gr}_{G, \leq \mu}^{\mathrm{adm}} = \Omega^{n-1}$ is the familiar Drinfeld space. In this particular example, a classic calculation of Drinfeld for $n = 2$ and Schneider–Stuhler for all n (see [Dri74] and [SS91], respectively) yields an isomorphism

$$R\Gamma_c(\Omega^{n-1}, \overline{\mathbb{Q}}_\ell[n-1]) \simeq \mathrm{St} \oplus \pi_{\{2,3,\dots,n-1\}}[-1] \oplus \dots \oplus \pi_{\{n-1\}}[2-n] \oplus \mathbf{1}[1-n]$$

as smooth $G(E)$ -representations. For general minuscule μ , the admissible locus $\mathrm{Gr}_{G, \leq \mu}^{\mathrm{adm}}$ is an example of a p -adic period domain in the sense of [RZ96]. However, in all cases beyond Drinfeld space, the question of computing the cohomology of the admissible locus has remained completely open. To the best of our knowledge, there was not even a precise conjecture.

Theorem 1.3.1. *For any μ , the cohomology $R\Gamma_c(\mathrm{Gr}_{G, \leq \mu}^{\mathrm{adm}}, \mathrm{IC}_\mu)$ is a direct sum of shifted irreducible $G(E)$ -representations, of total length $\dim V_\mu$. Moreover, there is an effective algorithm to compute this complex explicitly.*

We note that by general nonsense, the only irreducible representations which can appear here are the π_I 's, but essentially nothing else was known about this cohomology. In particular, the π_I 's admit many non-vanishing mutual Ext groups, so the fact that the cohomology complex is totally split is somewhat surprising. Even in the situation where $\mu = (1, \dots, 1, 0, \dots, 0)$, so $\mathrm{Gr}_{G, \leq \mu}$ is just the usual Grassmannian $\mathrm{Gr}(d, n)$, these results are totally new beyond the Drinfeld case.

The first really interesting examples are given as follows.

Example 1.3.2. In case of $\mathrm{Gr}(2, 5)$, our algorithm easily computes the cohomology of the admissible locus as

$$\begin{aligned} & \mathrm{St}^2 \oplus \pi_{\{2,3,4\}}[-1] \oplus \pi_{\{1,2,4\}}[-1] \oplus \pi_{\{1,2\}}[-2] \oplus \pi_{\{2,4\}}[-2] \\ & \oplus \pi_{\{2\}}[-3] \oplus \pi_{\{4\}}[-3] \oplus \mathbf{1}[-4] \oplus \mathbf{1}[-6]. \end{aligned}$$

Example 1.3.3. Similarly, for $\mathrm{Gr}(3, 6)$, the cohomology is

$$\begin{aligned} & \mathrm{St}[2]^2 \oplus \pi_{\{1,2,3,5\}}[1] \oplus \pi_{\{1,3,4,5\}}[1] \\ & \oplus \pi_{\{1,2,3\}} \oplus \pi_{\{1,3,5\}} \oplus \pi_{\{3,4,5\}} \\ & \oplus \pi_{\{1,3\}}[-1] \oplus \pi_{\{1,5\}}[-1] \oplus \pi_{\{3,5\}}[-1] \\ & \oplus \pi_{\{1\}}[-2] \oplus \pi_{\{1\}}[-4] \oplus \pi_{\{3\}}[-2] \oplus \pi_{\{5\}}[-2] \oplus \pi_{\{5\}}[-4] \\ & \oplus \mathbf{1}[-3] \oplus \mathbf{1}[-5]^2 \oplus \mathbf{1}[-7] \oplus \mathbf{1}[-9]. \end{aligned}$$

Both two examples above are deduced from the algorithm claimed in Theorem 1.3.1. The starting point for this result is the isomorphism

$$R\Gamma_c(\mathrm{Gr}_{G, \leq \mu}^{\mathrm{bas}}, \mathrm{IC}_\mu) \simeq i_1^* T_{V_\mu^\vee} i_{b_{d/n}}! \mathbf{1},$$

which is a simple consequence of the Fargues–Scholze machinery. In fact, we can algorithmically compute any composition $i_{b_{d/n}}^* T_{V_\mu} i_{b_{e/n}}! \rho$ where ρ is any generalized Steinberg of G_{b_e} and $\mu = (w_1, \dots, w_n)$ is any dominant weight. Note that this composition vanishes unless $d = e + \sum_{1 \leq i \leq n} w_i$. When it does not vanish, the algorithm is the following.

Step 1. Using Conjecture 1.1.4, write down the basic weight λ and shift s such that

$$i_{b_{e/n}}! \rho = \mathcal{L}_\lambda^{\mathrm{aut}}[s].$$

Step 2. Since $V_\mu \otimes \mathcal{L}_\lambda = \bigoplus_\nu \mathcal{L}_{\lambda+\nu}^{\dim V_\mu(\nu)}$, categorical Langlands implies that

$$T_{V_\mu} i_{b_{e/n}}! \rho = \bigoplus_\nu \mathcal{L}_{\lambda+\nu}^{\mathrm{aut}}[s]^{\dim V_\mu(\nu)},$$

where λ and s are chosen as in Step 1.

Step 3. From Step 2, we formally get

$$i_{b_{d/n}}^* T_{V_\mu} i_{b_{e/n}}! \rho = \bigoplus_\nu i_{b_{d/n}}^* \mathcal{L}_{\lambda+\nu}^{\mathrm{aut}}[s]^{\dim V_\mu(\nu)},$$

and now the stalks $i_{b_{d/n}}^* \mathcal{L}_{\lambda+\nu}^{\mathrm{aut}}$ can be read off from Conjecture 1.1.2.

Note that this algorithm *proves* Theorem 1.3.1, as the individual stalks $i_{b_{d/n}}^* \mathcal{L}_{\lambda+\nu}^{\text{aut}}$ are shifted irreducible representations, and the direct sum appearing in Step 3 clearly contributes a totality of $\dim V_\mu$ such stalks.

The fun here is in the examples. Here are a few more.

Example 1.3.4. On GL_6 ,

$$i_{b_{3/6}}^* T_{\wedge^3 \text{std}} i_{1!} \text{St} = \text{St}^{14} \oplus \rho_{\{1\}}[-1]^2 \oplus \rho_{\{2\}}[-1]^2 \\ \oplus \mathbf{1}[-2] \oplus \mathbf{1}[-4]$$

and

$$i_{b_{3/6}}^* T_{\wedge^3 \text{std}} i_{1!} \mathbf{1} = \text{St}[5]^2 \oplus \rho_{\{1\}}[4] \oplus \rho_{\{2\}}[4] \oplus \rho_{\{1\}}[2] \oplus \rho_{\{2\}}[2] \\ \oplus \mathbf{1}[3] \oplus \mathbf{1}[1]^3 \oplus \mathbf{1}[-1]^3 \oplus \mathbf{1}[-3]^3 \oplus \mathbf{1}[-5]^2 \oplus \mathbf{1}[-7] \oplus \mathbf{1}[-9]$$

and

$$i_1^* T_{\wedge^3 \text{std} \vee i_{b_{3/6}}!} \text{St} = \text{St}^5 \oplus \pi_{\{1,2,3,4\}}[-1]^2 \oplus \pi_{\{2,3,4,5\}}[-1]^2 \oplus \pi_{\{1,2,4,5\}}[-1] \\ \oplus \pi_{\{1,2,4\}}[-2] \oplus \pi_{\{2,3,4\}}[-2] \oplus \pi_{\{2,4,5\}}[-2] \\ \oplus \pi_{\{1,2\}}[-3] \oplus \pi_{\{2,4\}}[-3] \oplus \pi_{\{4,5\}}[-3] \\ \oplus \pi_{\{2\}}[-4] \oplus \pi_{\{4\}}[-4] \\ \oplus \mathbf{1}[-5] \oplus \mathbf{1}[-7].$$

Example 1.3.5. On GL_8 ,

$$i_{b_{4/8}}^* T_{\text{std}} i_{b_{3/8}!} \mathbf{1} = \text{St}^2 \oplus \rho_{\{1,2\}}[-1]^2 \oplus \rho_{\{1\}}[-2] \oplus \rho_{\{1\}}[-4] \oplus \mathbf{1}[-5] \oplus \mathbf{1}[-7]$$

and

$$i_{b_{4/8}}^* T_{\wedge^2 \text{std}} i_{b_{2/8}!} \text{St} = \text{St}^9 \oplus \rho_{\{2,3\}}[-1]^2 \oplus \rho_{\{1,2\}}[-1]^3 \oplus \rho_{\{2,3\}}[-3]^2 \oplus \rho_{\{1,2\}}[-3]^2 \\ \oplus \rho_{\{2\}}[-2] \oplus \rho_{\{2\}}[-4]^2 \oplus \rho_{\{1\}}[-4] \oplus \rho_{\{2\}}[-6] \\ \oplus \mathbf{1}[-5] \oplus \mathbf{1}[-7]^2 \oplus \mathbf{1}[-9] \oplus \mathbf{1}[-11]$$

and

$$i_{b_{2/8}}^* T_{\wedge^2 \text{std}} i_{1!} \text{St} = \text{St}^{22} \oplus \mathbf{1}[-1]^3 \oplus \mathbf{1}[-3]^2 \oplus \mathbf{1}[-5]$$

and

$$i_{b_{4/8}}^* T_{\wedge^4 \text{std}} i_{1!} \text{St} = \text{St}^{33} \oplus \text{St}[-2]^9 \\ \oplus \rho_{\{1,2\}}[-1]^5 \oplus \rho_{\{1,3\}}[-1]^4 \oplus \rho_{\{2,3\}}[-1]^4 \oplus \rho_{\{2,3\}}[-3] \\ \oplus \rho_{\{2\}}[-2] \oplus \rho_{\{1\}}[-2]^2 \oplus \rho_{\{1\}}[-4]^2 \oplus \rho_{\{3\}}[-2]^2 \oplus \rho_{\{3\}}[-4]^2 \\ \oplus \mathbf{1}[-3] \oplus \mathbf{1}[-5]^2 \oplus \mathbf{1}[-7] \oplus \mathbf{1}[-9].$$

Example 1.3.6. On GL_9 ,

$$i_{b_{3/9}}^* T_{\text{std}} i_{b_{2/9}!} \mathbf{1} = \text{St}^3 \oplus \rho_{\{1\}}[-1] \oplus \rho_{\{1\}}[-3]^2 \\ \oplus \mathbf{1}[-4] \oplus \mathbf{1}[-6] \oplus \mathbf{1}[-8].$$

1.4. Canonical extensions. In the course of our analysis at non-split points, we actually compute all the ext-groups between the sheaves $\mathcal{L}_\lambda$.

Theorem 1.4.1. *For any weights $\mu, \lambda \in \mathbb{X}^*(\hat{T})$, we have $R\mathrm{Hom}(\mathcal{L}_\lambda, \mathcal{L}_\mu) = 0$ unless $\mu = \lambda + \sum_{1 \leq i \leq n-1} m_i \alpha_i$, and in the non-vanishing case*

$$R\mathrm{Hom}(\mathcal{L}_\lambda, \mathcal{L}_\mu) = k[-\sum_{i \in I^c} 2m_i] \oplus k[-1 - \sum_{i \in I^c} 2m_i],$$

where $I = \{1 \leq i \leq n-1 : m_i \leq 0\}$ as before.

The class in odd degree here is “parasitic”, and would vanish if we were working with PGL_n instead of GL_n . However, the class in even degree is much more interesting. Note that in the case where $\mu = \lambda \pm \alpha$ for some simple α , the even degree class corresponds to a canonical map $L_\alpha : \mathcal{L}_\lambda \rightarrow \mathcal{L}_{\lambda-\alpha}$, resp. a canonical map $R_\alpha : \mathcal{L}_\lambda \rightarrow \mathcal{L}_{\lambda+\alpha}[2]$. Following a suggestion of Bertoloni Meli, we call these classes *lowering* and *raising* operators. It is not hard to show that the even degree class in Theorem 1.4.1 is uniquely a composition of iterated raising and lowering operators.

We now consider the following construction. Fix $I \subseteq \{1, \dots, n-1\}$, and set

$$\begin{aligned} \mathcal{F}_I &= \operatorname{colim}_{\min m_i \rightarrow \infty} \mathcal{L}_{-\sum_{i \in I} m_i \alpha_i + \sum_{i \in I^c} m_i \alpha_i}[-|I^c| + 2 \sum_{i \in I^c} m_i] \\ &\in \mathrm{IndCoh}(\mathrm{Par}_{\mathrm{GL}_n}) \end{aligned}$$

where the colimit is taken along the obvious compositions of raising (in I^c) and lowering (in I) operators. Note that the shift is equivalently $\sum_{i \in I^c} (2m_i - 1)$.

Let $\mathcal{F}_I^{\mathrm{aut}}$ be the associated automorphic sheaf under CLLC. Note that by construction, $i_1^* \mathcal{F}_I^{\mathrm{aut}} = \pi_I$, so the sheaf $\mathcal{F}_I^{\mathrm{aut}}$ is some preferred extension of π_I from the point $b = 1$. This extension turns out to have some very interesting properties.

Proposition 1.4.2. *The sheaf $\mathcal{F}_I^{\mathrm{aut}}$ is ULA with all stalks of finite length, and indecomposable in the sense that $\mathrm{End}(\mathcal{F}_I^{\mathrm{aut}}) = k$. Moreover, $\mathcal{F}_I^{\mathrm{aut}}$ is a sheared Hecke eigensheaf: if V is any degree zero irreducible representation, then $T_V \mathcal{F}_I^{\mathrm{aut}}$ is a direct sum of shifts of $\mathcal{F}_I^{\mathrm{aut}}$.*

If b is any point in the closure of $b = 1$ with integral Newton slopes, so G_b is a Levi subgroup of G , then in the Grothendieck group we have $[i_b^{\mathrm{ren}} \mathcal{F}_I^{\mathrm{aut}}] = [r_G^{P^-} \pi_I]$ and $[i_b^{\mathrm{lren}} \mathcal{F}_I^{\mathrm{aut}}] = [r_G^P \pi_I]$.*

We also conjecture that $\mathbb{D}_{\mathrm{Verd}} \mathcal{F}_I^{\mathrm{aut}} = \mathcal{F}_I^{\mathrm{aut}}$.

1.5. Notations and conventions.

1.5.1. General conventions about integers.

- In the monoid of non-negative integers $\mathbb{N}$, for each element $n \geq 2$, denote by $[n-1]$ the ordered subset $\{1, \dots, n-1\} \subseteq \mathbb{N}$.
- Fix two integers n and d . Let (d, n) be their greatest common divisor. Write $n' := n/(d, n)$ and $d' := d/(d, n)$.
- For n and d fixed as above, define the symbol $\delta_i \in \{0, 1\}$ for each $i \in [n-1]$ by setting

$$\delta_i := \begin{cases} 1, & \text{if } n' \mid i, \\ 0, & \text{otherwise.} \end{cases}$$

- Fix integer $n \geq 2$. Denote by $\mathbf{r}$ the $(n-1)$ -tuple $(r_1, \dots, r_{n-1}) \in \mathbb{Z}^{n-1}$. Depending on the context, if $\mathbf{1}$ does not denote the trivial representation of any group, it will be the $(n-1)$ -tuple $(1, \dots, 1) \in \mathbb{Z}^{n-1}$.

1.5.2. General conventions in category theory.

- Given any functor F , we use the symbol $(-)$ to denote the virtual test object putting inside; so we also write $F = F(-)$.
- Given an object in some category, we use the symbol $[-]$ to denote the corresponding virtual object in the Grothendieck group.

1.5.3. Groups and representations.

Let $G = \mathrm{GL}_n$ for $n \geq 2$ over a finite extension E of $\mathbb{Q}_p$.

- Let T be the usual maximal split torus of G , and then $\mathbb{X}^*(\hat{T}) = \mathbb{Z}^n$.
- For each $1 \leq i \leq n-1$, let $\alpha_i := \varepsilon_i - \varepsilon_{i+1} \in \mathbb{X}^*(\hat{T})$ be the simple root.
- Given a subset $I \subseteq [n-1]$, let P_I be the associated standard parabolic with Levi M_I .

For the last term above, recall that the standard parabolics in GL_n can be parametrized via subsets $I \subseteq [n-1]$. Explicitly, if we write elements in I as $i_1 < \dots < i_k$, then P_I be the standard block upper triangular parabolic with Levi $M_I = \mathrm{GL}_{i_1} \times \mathrm{GL}_{i_2-i_1} \times \dots \times \mathrm{GL}_{i_k-i_{k-1}} \times \mathrm{GL}_{n-i_k}$.

Moreover, consider the following ingredients about the tori.

- Given $\sigma \in S_n$, let $w_\sigma = (\delta_{i,\sigma(j)})$ be the associated permutation matrix of size $n \times n$.

We identify S_n with the evident subgroup of GL_n via the map $\sigma \rightarrow w_\sigma$. One easily checks that on each element $t = \mathrm{diag}(t_1, \dots, t_n)$, the S_n -action by σ is given as $w_\sigma^{-1} t w_\sigma = \mathrm{diag}(t_{\sigma(1)}, \dots, t_{\sigma(n)})$. Also, for $\sigma \in S_n$, conjugation by w_σ induces actions both on $\mathbb{X}^*(\hat{T})$ and on smooth characters of $T(E)$, defined by $(\sigma\chi)(t) = \chi(w_\sigma^{-1} t w_\sigma)$.

- Let φ be the semisimple L-parameter which is trivial on inertia and sends geometric Frobenius to $\mathrm{diag}(q^{\frac{1-n}{2}}, \dots, q^{\frac{n-1}{2}})$.
- Let $\delta^{1/2}$ be the unramified character of $T(E)$ sending $t = \mathrm{diag}(t_1, \dots, t_n)$ to $\prod_{1 \leq i \leq n} |t_i|^{\frac{n+1-2i}{2}}$.

Note that under classical LLC for tori, $\delta^{1/2}$ matches φ , and $\sigma\delta^{1/2}$ matches $\sigma\varphi := w_\sigma\varphi w_\sigma^{-1}$.

More notations on local smooth representation theory will be set in Subsection 2.1.

1.5.4. Some notations for weights.

- Suppose $\lambda = (l_1, \dots, l_n) \in \mathbb{X}^*(\hat{T}) = \mathbb{Z}^n$ is a weight of degree d , i.e., $l_1 + \dots + l_n = d$. Then there are integers $m_1, \dots, m_{n-1} \in \mathbb{Z}$, uniquely determined by λ , such that

$$\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i,$$

where $\omega_d = (\lceil \frac{d}{n} \rceil, \lceil \frac{2d}{n} \rceil - \lceil \frac{d}{n} \rceil, \lceil \frac{3d}{n} \rceil - \lceil \frac{2d}{n} \rceil, \dots, d - \lceil \frac{(n-1)d}{n} \rceil)$, the Steinberg weight vector of G_{b_d} , is of degree d and independent of λ . Note that $\sum_{1 \leq i \leq n-1} m_i \alpha_i$ is a weight of degree zero.

- Given an $(n-1)$ -tuple $\mathbf{k} = (k_1, \dots, k_{n-1}) \in \mathbb{Z}^{n-1}$, for any subset $I \subseteq [n-1]$, the associated weight $\lambda_{I,\mathbf{k}}$ of degree zero with coefficient $\mathbf{k}$ is the summation

$$\lambda_{I,\mathbf{k}} := - \sum_{i \in I} k_i \alpha_i + \sum_{i \in I^c} k_i \alpha_i,$$

with $\alpha_i \in \mathbb{X}^*(\hat{T})$ the prescribed simple roots. Also define the transpose of $\lambda_{I,\mathbf{k}}$ by

$$\lambda_{I,\mathbf{k}}^\dagger := - \sum_{i \in I} k_i \alpha_{n-i} + \sum_{i \in I^c} k_i \alpha_{n-i}.$$

If we write $\mathbf{k}^t = (k_n, \dots, k_1)$, then we have $\lambda_{I,\mathbf{k}}^\dagger = \lambda_{I^t,\mathbf{k}^t}$.

1.5.5. *Some subsets of $\{1, \dots, n-1\}$.*

- Given any weight $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i \in \mathbb{X}^*(\hat{T})$ of degree d , with $m_i \in \mathbb{Z}$, define the set

$$I_\lambda := \{i \in [(d, n) - 1] : m_{n'-i} \leq 0\} \subseteq [(d, n) - 1].$$

Here $n' = n/(d, n)$ as above. Throughout the paper, the notation I_λ will be mostly used for those λ of degree zero, in which case this definition simplifies into

$$I_\lambda := \{i \in [n-1] : m_i \leq 0\} \subseteq [n-1].$$

Unless otherwise stated, the weight λ has been already chosen, so the integers m_i 's are fixed.

- For an arbitrary subset $I \subseteq [n-1]$, define the *complement* and the *transpose* of I , denoted by I^c and I^t respectively, as

$$I^c := [n-1] \setminus I, \quad I^t := \{n-i : i \in I\}.$$

Both of I^c and I^t are subsets of $[n-1]$ as well.

- For each $\sigma \in S_n$, where S_n is the finite Weyl group of $G = \mathrm{GL}_n$, define the associated *ascent set* as

$$\mathrm{Asc}(\sigma) := \{j \in [n-1] : \sigma(j) < \sigma(j+1)\}.$$

From the construction, we notice that $I^{\mathrm{ct}} = I^c$. Also, another simple observation is that $\mathrm{Asc}(w_0\sigma) = \mathrm{Asc}(\sigma)^c$, where $(w_0 : j \mapsto n+1-j) \in S_n$ is the longest element in the finite Weyl group for GL_n .

2. REPRESENTATION THEORY

2.1. Generalized Steinberg representations. In this section, we establish some basic facts and notations concerning the irreducible smooth $\mathrm{GL}_n(E)$ -representations with semisimple L-parameter φ .

We begin with a reminder on dualities. Smooth duality $(-)^{\vee}$ is t-anti-exact, preserves admissible representations, commutes with i_P^G , and intertwines r_G^P with $r_G^{P^-}$. Bernstein's cohomological duality $\mathbb{D}_{\mathrm{coh}}$ is defined on $D(G(E), \Lambda)^\omega = D_{\mathrm{fg}}^b(G(E), \Lambda)$, commutes with r_G^P , and intertwines i_P^G with $i_{P^-}^G$. It is not t-anti-exact, but on finite-length objects it is so up to an explicit shift: if π is an irreducible representation, then

$$\mathbb{D}_{\mathrm{coh}}\pi = \pi'[-d]$$

for some irreducible π' , where d is the dimension of the Bernstein component containing π . This suggests the following construction: writing $e_{\mathfrak{s}}$ for the primitive idempotents in the Bernstein center, and $d_{\mathfrak{s}}$ for the dimension of the associated components, we set

$$\mathbb{D}_{\text{ncoh}} A = \bigoplus_{\mathfrak{s}} \mathbb{D}_{\text{coh}}(e_{\mathfrak{s}} A)[d_{\mathfrak{s}}].$$

Then $\mathbb{D}_{\text{ncoh}}$ is still an involutive anti-equivalence on $\mathcal{D}_{\text{fg}}^b(G(E), \Lambda)$, and on the subcategory $\mathcal{D}(G(E), \Lambda)_{\text{fin}}$ of objects whose cohomology is of finite length, it is t -anti-exact. Finally, we set $\text{Zel}(A) = (\mathbb{D}_{\text{ncoh}} A)^{\vee}$. This is a *covariant t -exact* self-equivalence of $\mathcal{D}(G(E), \Lambda)_{\text{fin}}$, which then restricts to a covariant self-equivalence on the heart $\text{Rep}(G(E), \Lambda)_{\text{fin}}$. It's easy to see that $\text{Zel}(\sigma) = \sigma$ for any supercuspidal σ , and $\text{Zel}(\nu\sigma) = \nu\text{Zel}(\sigma)$ for any character ν . It's also easy to see that Zel intertwines r_G^P with r_G^{P-} and i_P^G with i_{P-}^G .

Now return to $G = \text{GL}_n$, and let $P_I \subseteq G$ be the parabolic associated with some $I = \{i_1 < i_2 < \dots < i_k\} \subseteq [n-1]$ as in 1.5.3. Consider $\mathcal{C}(P_I(E) \backslash G(E)) = i_{P_I}^G(\delta_{P_I}^{-1/2})$. The following proposition on Jacquet modules has been essentially detected by Zelevinsky; here, we rephrase and renormalize the results in [Zel80, §2] into our context.

Proposition 2.1.1. *We have a direct decomposition of Jacquet modules*

$$r_G^B i_{P_I}^G(\delta_{P_I}^{-1/2}) = \bigoplus_{\sigma: \text{Asc}(\sigma) \subseteq I^t} \sigma \delta^{1/2}.$$

Here, recall that $\text{Asc}(\sigma) = \{j \in [n-1]: \sigma(j) < \sigma(j+1)\}$ is the ascent set of the permutation $\sigma \in S_n$.

Proof. By the geometric lemma, we know that $r_G^B i_{P_I}^G$ is glued from the functors $\sigma \circ r_{M_I}^{B \cap M_I}$, where σ runs over elements such that $\sigma(j) < \sigma(l)$ for all integers such that $j < l$ and both j, l belong to the same of the blocks $\{1, 2, \dots, i_1\}, \{i_1 + 1, \dots, i_2\}, \{i_2 + 1, \dots, i_3\}, \dots, \{i_k + 1, \dots, n\}$. Clearly, σ satisfies this condition if and only if $\sigma(j) < \sigma(j+1)$ for all of those j such that j and $j+1$ belong to the same block. Asking that j and $j+1$ belong to the same block is the same as requiring that $i_m + 1 \leq j$ and $j+1 \leq i_{m+1}$ for some m , which is equivalent to $j \in I^c$. In other words, the relevant σ 's are exactly those with $I^c \subseteq \text{Asc}(\sigma)$. Thus, $r_G^B i_{P_I}^G$ is glued from the functors $\sigma \circ r_{M_I}^{B \cap M_I}$, where σ runs over all elements such that $I^c \subseteq \text{Asc}(\sigma)$.

Now applying this to the M_I -representation $\delta_{P_I}^{-1/2}$, it is easy to see that

$$r_{M_I}^{B \cap M_I}(\delta_{P_I}^{-1/2}) = \delta^{-1/2} = w_0 \delta^{1/2}.$$

Therefore,

$$r_G^B i_{P_I}^G(\delta_{P_I}^{-1/2}) = \bigoplus_{\sigma: \text{Asc}(\sigma) \supseteq I^c} \sigma w_0 \delta^{1/2} = \bigoplus_{\sigma: \text{Asc}(\sigma w_0) \supseteq I^c} \sigma \delta^{1/2}.$$

But it is also easy to check that $\text{Asc}(\sigma w_0) = \text{Asc}(\sigma)^{\text{ct}}$, so the condition $\text{Asc}(\sigma w_0) \supseteq I^c$ is equivalent to the condition $\text{Asc}(\sigma) \subseteq I^t$. This gives the result. $\square$

Let π_I be the unique irreducible quotient of the space $\mathcal{C}(P_I^-(E) \backslash G(E))$, i.e.,

$$\mathcal{C}(P_I^-(E) \backslash G(E)) \simeq i_{P_I^-}^G(\delta_{P_I^-}^{-1/2}) \simeq i_{P_{I^t}}^G(\delta_{P_{I^t}}^{-1/2}) \twoheadrightarrow \pi_I.$$

For varying I , the π_I 's are exactly the irreducible subquotients of $i_B^G(\sigma\delta^{1/2})$ (for every choice of σ), and they are exactly the irreducible $G(E)$ -representations with semisimple parameter φ .

Proposition 2.1.2. *The generalized Steinberg representation π_I admits the following properties.*

(1) *The Jacquet module of π_I admits a direct decomposition*

$$r_G^B(\pi_I) = \bigoplus_{\sigma: \text{Asc}(\sigma)=I} \sigma\delta^{1/2}.$$

(2) *For each subset I , the smooth duality, Bernstein's cohomological duality, and Zelevinsky duality of π_I are respectively given by $\pi_I^\vee = \pi_{I^t}$, $\mathbb{D}_{\text{coh}}\pi_I = \pi_{I^c}[1-n]$, and $\text{Zel}(\pi_I) = \pi_{I^c}$.*

(3) *The standard module of π_I , denoted by π_I^{mod} , is given by*

$$\pi_I^{\text{mod}} = i_{P_{I^c}}^G(\delta_{P_{I^c}}^{1/2} \cdot \text{St}_{M_{I^c}}).$$

(4) *The L -parameter of π_I is (φ, N_I) where $N_I = \sum_{i \in I} E_{i,i+1}$. Here $E_{i,j}$ is the usual elementary matrix with a 1 in the (i,j) -spot and all other entries 0.*

Note that Proposition 2.1.2 (4) above accounts for our choice of normalization of $I \rightsquigarrow \pi_I$.

Proof. (1) By general facts, $\mathcal{C}(P_I^-(E) \backslash G(E))$ is multiplicity-free and $\ker(\mathcal{C}(P_I^-(E) \backslash G(E)) \rightarrow \pi_I)$ contains π_J as a subquotient for all $J \subsetneq I$. The first part then follows from the previous proposition by ascending induction on $|I|$.

(2) This follows easily from part (1) using the known interactions of these operations with r_G^B as recalled above. For example, to determine $\pi_I^\vee$, we compute

$$\bigoplus_{\sigma: \text{Asc}(\sigma)=I} \sigma\delta^{-1/2} = (r_G^B(\pi_I))^\vee \simeq r_G^{B^-}(\pi_I^\vee) \simeq w_0 \circ r_G^B(\pi_I^\vee)$$

where the first and third equalities are formal. Since also $\sigma\delta^{-1/2} = \sigma w_0 \delta^{1/2}$, this gives

$$r_G^B(\pi_I^\vee) = \bigoplus_{\sigma: \text{Asc}(\sigma)=I} w_0 \sigma w_0 \delta^{1/2} = \bigoplus_{\sigma: \text{Asc}(\sigma)=I^t} \sigma\delta^{1/2}$$

where the second equality follows from the easy observation that $\text{Asc}(w_0 \sigma w_0) = \text{Asc}(\sigma)^t$. Since $\pi_I^\vee = \pi_J$ for some J , this shows that $J = I^t$ and thus $\pi_I^\vee = \pi_{I^t}$. Similar arguments dispense with $\mathbb{D}_{\text{coh}}\pi_I$ and $\text{Zel}(\pi_I)$.

(3) Apply $\text{Zel}(-)$ to the surjection

$$i_{P_I^-}^G(\delta_{P_I^-}^{-1/2}) \twoheadrightarrow \pi_I.$$

Using the basic properties of Zel recalled above, together with the fact that $\text{Zel}(\mathbf{1}_{M_I}) = \text{St}_{M_I}$ and $\delta_{P_I^-}^{-1/2} = \delta_{P_I}^{1/2}$, we get a surjection

$$i_{P_I}^G(\delta_{P_I}^{1/2} \text{St}_{M_I}) = i_{P_I}^G(\delta_{P_I^-}^{-1/2} \text{St}_{M_I}) = \text{Zel}(i_{P_I^-}^G(\delta_{P_I^-}^{-1/2})) \twoheadrightarrow \text{Zel}(\pi_I) = \pi_{I^c}$$

where the final identity follows from the second part. But the source is clearly a standard module, since St_{M_I} is tempered and $\delta_{P_I}^{1/2}$ is P_I -positive. Replacing I by I^c gives the third part.

(4) It is enough to compute the L-parameter of $\delta_{P_{\text{fc}}}^{1/2} \text{St}_{M_{\text{fc}}}$, since by definition the Langlands quotient π of a standard module $i_P^G(\nu \cdot \sigma)$ satisfies $\phi_\pi = \nu \cdot \phi_\sigma$. This is straightforward and left to the reader. $\square$

These arguments also give some other useful facts along the way, e.g. that π_I^{mod} is multiplicity-free with Jordan–Holder factors π_J for all $I \subseteq J$.

Proposition 2.1.3. *Under CLLC, the standard module π_I^{mod} goes to the structure sheaf of the substack of V_φ defined by setting $v_i = 0$ for all $i \in I^c$.*

2.2. Categorical local Langlands. In this section we briefly recall what we need from the categorical local Langlands correspondence. For the moment, we allow G to be any quasisplit group over E , with a fixed Whittaker datum (B, ψ) . Let Bun_G be the stack of G -bundles on the Fargues–Fontaine curve, and let Par_G be the stack of ℓ -adically continuous L-parameters into ${}^L G$. By the main result of Fargues–Scholze, we have the *spectral action* of $\text{QCoh}(\text{Par}_G)$ on $\text{D}(\text{Bun}_G)$. Let $a_\psi: \text{QCoh}(\text{Par}_G) \rightarrow \text{D}(\text{Bun}_G)$ be the functor induced by acting on $i_{1!} W_\psi$, and let c_ψ be its right adjoint. The categorical local Langlands conjecture is the following statement.

Conjecture 2.2.1. *There is a (necessarily unique) equivalence of categories*

$$\mathbf{L}_\psi: \text{D}(\text{Bun}_G) \xrightarrow{\sim} \text{IndCoh}(\text{Par}_G)$$

such that $c_\psi = \Psi \circ \mathbf{L}_\psi$, where $\Psi: \text{IndCoh}(\text{Par}_G) \rightarrow \text{QCoh}(\text{Par}_G)$ is the usual quotient functor.

In practice we will need a little bit more. For convenience, we refer to the following statement as *strong categorical local Langlands*.

Conjecture 2.2.2. *There is a (necessarily unique) equivalence of categories*

$$\mathbf{L}_\psi: \text{D}(\text{Bun}_G) \xrightarrow{\sim} \text{IndCoh}(\text{Par}_G)$$

such that $c_\psi = \Psi \circ \mathbf{L}_\psi$ for G and for all standard Levi subgroups of G . Moreover, the equivalence $\mathbf{L}_\psi$ satisfies the following compatibility properties.

- (1) *On compact sheaves, we have $\mathbb{D}_{\text{tw,GS}} \mathbf{L}_\psi \simeq \mathbf{L}_{\psi^{-1}} \mathbb{D}_{\text{BZ}}$.*
- (2) *For any standard parabolic $P = MU \subseteq G$, there is an equivalence $\mathbf{L}_\psi \text{Eis}_{P!} \simeq \text{Eis}_{P-}^{\text{spec}} \mathbf{L}_{\psi_M}$.*

One of the main results of [HM26] is the following theorem.

Theorem 2.2.3. *The following are equivalent.*

- (1) *The strong categorical local Langlands conjecture is true for GL_n , for all n .*
- (2) *There is an equivalence of functors*

$$c_\psi \text{Eis}_{P!} \simeq \text{Eis}_{P-}^{\text{spec,coarse}} c_{\psi_M}: \text{D}(\text{Bun}_M)^\omega \longrightarrow \text{QCoh}(\text{Par}_G),$$

where $G = \text{GL}_n$ and $M \subseteq G$ is any standard Levi.

As discussed in [HM26], the second condition here should be verified in the near future (uniformly in all groups).

In much of this paper, we assume strong categorical local Langlands for GL_n . Since any such group admits a unique Whittaker datum, we write $\mathbf{L}_G$ for the Langlands functor in this case rather than $\mathbf{L}_\psi$.

2.3. Vogan sheaves: first results. In this section, we assume strong categorical local Langlands for $G = \mathrm{GL}_n$. Let $\mathcal{L}_\lambda$ be the family of coherent sheaves defined in the introduction, and let

$$\mathcal{L}_\lambda^{\mathrm{aut}} := \mathbf{L}_G^{-1} \mathcal{L}_\lambda$$

be the associated family of automorphic sheaves on Bun_G . In this section we give some first results on these sheaves.

Proposition 2.3.1. *For any weight λ of degree d , the automorphic sheaf $\mathcal{L}_\lambda^{\mathrm{aut}}$ has quasicompact support, with all stalks of finite length, and it is supported on the degree d connected component of Bun_G . Any irreducible subquotient of $i_{b_d}^* \mathcal{L}_\lambda^{\mathrm{aut}}$ is a generalized Steinberg representation of $G_{b_d}(E)$.*

Proof. Quasicompact support and finite length stalks are equivalent to the fact that $\mathcal{L}_\lambda$ is a finite coherent sheaf. The claim about which component supports the sheaf is equivalent to the fact that $\mathcal{L}_\lambda$ is isotypic for the degree d character of $Z(\hat{G})^\Gamma = \mathbb{G}_m$. The final claim is formal nonsense, since any such irreducible subquotient necessarily has Fargues–Scholze parameter φ , and Fargues–Scholze parameters agree with semisimple L-parameters for inner forms of GL_n by [HKW22]. $\square$

Recall from the introduction that a degree d weight $\omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ is *basic* if $m_i \in \{0, 1\}$ and $m_i = 0$ for all $n' \nmid i$. Here $n' = n/(d, n)$ as usual.

Proposition 2.3.2. *For any degree d weight λ , the automorphic sheaf $\mathcal{L}_\lambda^{\mathrm{aut}}$ is supported at the point b_d if and only if λ is a basic weight.*

Proof. This follows from work of Bertoloni Meli–Koshikawa. More precisely, there is a notion of basic coherent sheaf on Par_G , and under strong categorical local Langlands an automorphic sheaf A is basic if and only if $\mathbf{L}_G A$ is basic. Now Bertoloni Meli–Koshikawa have proved that $\mathcal{L}_\lambda$ is basic if and only if λ is a basic weight in the above sense. This gives the result. $\square$

Let $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ be any weight of degree d . Then we obtain its dual weight

$$\lambda^* = \omega_d + \sum_{1 \leq i \leq n-1} (\delta_i - m_{n-i}) \alpha_i$$

where $\delta_i \in \{0, 1\}$ and $\delta_i = 1$ if and only if $n' \mid i$. Note that $\lambda \mapsto \lambda^*$ preserves the degree, so λ^* is of degree d as well.

Lemma 2.3.3. *The assignment $\lambda \mapsto \lambda^*$ preserves basic weights.*

Proof. Recall from Subsection 1.1 that, for basic $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$, we have $m_i \in \{0, 1\}$ and $m_i = 0$ unless $n' = n/(d, n) \mid i$. If $n' \nmid i$, then also $n' \nmid (n - i)$, hence $\delta_i = 0$ and $m_{n-i} = 0$, so the i -th coefficient of λ^* is 0. If $n' \mid i$, then $n' \mid (n - i)$, so $\delta_i = 1$ and $m_{n-i} \in \{0, 1\}$, hence the i -th coefficient of λ^* is $1 - m_{n-i} \in \{0, 1\}$. Therefore, λ^* is again basic. $\square$

Proposition 2.3.4. *For any weight λ with its dual weight λ^* , we have*

$$\mathbb{D}_{\mathrm{BZ}} \mathcal{L}_\lambda^{\mathrm{aut}} \simeq \mathcal{L}_{\lambda^*}^{\mathrm{aut}}[-1].$$

Proof. By the duality compatibility in strong categorical local Langlands, this is equivalent to the claim that

$$\mathbb{D}_{\mathrm{tw.GS}} \mathcal{L}_\lambda \simeq \mathcal{L}_{\lambda^*}[-1].$$

This can be shown using the following direct computation. More precisely, let $\lambda \mapsto \lambda'$ be the involution on weights sending $(k_1, \dots, k_n)$ to $(k_n + 1, k_{n-1}, \dots, k_2, k_1 - 1)$. It is completely formal to prove that $\mathbb{D}_{\text{tw.GS}} \mathcal{L}_\lambda \simeq \mathcal{L}_{\lambda'}[-1]$, using $\mathcal{L}_\lambda = \iota_* \mathcal{O}(\lambda)$ together with the commutation of ι_* with Grothendieck–Serre duality and the fact that the dualizing sheaf of V_φ is the (-1) -shifted line bundle of weight $(1, 0, \dots, 0, -1) = \sum_{1 \leq i \leq n-1} \alpha_i$.

It is then enough to check that $\lambda' = \lambda^*$. Since $(\nu + \sum_i m_i \alpha_i)' = \nu' - \sum_i m_{n-i} \alpha_i$, this reduces to checking that $\omega'_d = \omega_d + \sum_i \delta_i \alpha_i$; but the latter follows directly from the definition of ω_d , the Steinberg weight vector of G_{b_d} . $\square$

Finally, we recall that for degree zero basic weights, the sheaves $\mathcal{L}_\lambda$ have been identified by Hellmann.

Proposition 2.3.5. *Let $I \subseteq \{1, 2, \dots, n-1\}$ be any subset, so $\alpha_{I^c} = \sum_{i \in I^c} \alpha_i$ is a basic weight of degree zero, and all such weights are obtained in this form. Then*

$$\mathcal{L}_{\alpha_{I^c}}^{\text{aut}} \simeq i_{1!} \pi_I[-|I^c|].$$

Proof. An equivalent statement is asserted in [Hel23, Remark 4.43]. Note that the functor $\mathbf{L}_{G^{i_1!}}$ agrees with the (hypothetical) functor considered by Hellmann, so his results apply. $\square$

We know from Lemma 2.3.3 that $\lambda \mapsto \lambda^*$ preserves the basic weights, so the previous Proposition 2.3.4 and Proposition 2.3.5 are compatible in the evident sense.

3. CALCULATIONS WITH COHERENT SHEAVES

Most of our analysis will take place on the spectral side of categorical local Langlands. Here we will compute using smart local models for the parameter stacks formally completed along the Vogan stack of Steinberg. Our first goal is to record these local models.

3.1. Formal completions. Fix $G = \text{GL}_n$ as above, and let φ be the semisimple Steinberg parameter. We write X for the formal completion of Par_G along V_φ .

Proposition 3.1.1. *There is a natural isomorphism*

$$X \simeq (\text{Spf } k[[u_1, \dots, u_n]][v_1, \dots, v_{n-1}]/(u_1 v_1, \dots, u_{n-1} v_{n-1}))/\hat{T}$$

where $\hat{T}$ acts trivially on the variables $u_1, \dots, u_n$ and acts on v_i through α_i .

We omit the proof of Proposition 3.1.1, since an essentially identical description appears in Hellmann; see [Hel23, (4.24)]. The reader may refer to another detailed proof by X. Yang in [Yan25, Proposition 3.12]. The only difference is that *loc. cit.* uses the opposite Borel instead. Heuristically, for each classical point of X , the variables u_i serve as semisimple coordinates for the formal deformation space at the fixed regular semisimple element $\varphi(\text{Fr})$, whereas the variables v_i serve as nilpotent coordinates. The relations $u_i v_i = 0$ reflect the condition $\text{ad}(\varphi(\text{Fr})) \cdot N = q^{-1}N$ (up to $\text{Cent}_{\hat{G}}(\varphi)$ -conjugacy), which cuts out the Vogan stack.

Inside X stated above, let X_I be the closed substack defined by conditions $v_i = 0$ for all $i \in I^c$. Let $\tilde{X}$ be the formal completion of Par_B along the preimage of V_φ , which is equivalently the fiber product $\text{Par}_B \times_{\text{Par}_G} X$. There is a canonical proper map $p: \tilde{X} \rightarrow X$.

Proposition 3.1.2. *The connected components of $\tilde{X}$ are canonically indexed by the finite Weyl group $W = S_n$. Moreover, if we denote by $\tilde{X}_\sigma$ the connected component corresponding to $\sigma \in S_n$, then the map $p: \tilde{X} \rightarrow X$ induces an isomorphism*

$$p_\sigma: \tilde{X}_\sigma \xrightarrow{\sim} X_{\text{Asc}(\sigma)}.$$

In other words, there is a closed immersion from $\tilde{X}_\sigma$ into X , factoring over an isomorphism onto $X_{\text{Asc}(\sigma)}$.

Proof. Note that $p: \tilde{X} \rightarrow X$ is the base change of $\text{Par}_B \rightarrow \text{Par}_G$ towards the formal neighborhood X . Write $R := k[[u_1, \dots, u_n]][v_1, \dots, v_{n-1}]/(u_1 v_1, \dots, u_{n-1} v_{n-1})$ for the coordinate ring in Proposition 3.1.1. Then for each $I \subseteq [n-1]$, we have

$$X_I \simeq (\text{Spf } R/(v_i)_{i \in I^c})/\hat{T}.$$

In particular, $X = X_\emptyset \simeq (\text{Spf } R)/\hat{T}$. By our setting, $\varphi(\text{Fr})$ is semisimple with distinct eigenvalues, so the Borel subgroups in $\hat{G}$ containing $\varphi(\text{Fr})$ form a finite discrete set canonically indexed by $W = S_n$; for each $\sigma \in S_n$, the corresponding Borel is $\hat{B}_\sigma := w_\sigma^{-1} \hat{B} w_\sigma$. We then write $\tilde{X}_\sigma$ the formal neighborhood of those points in $\tilde{X}$ whose reductions to Borels are exactly $\hat{B}_\sigma$; note that such $\tilde{X}_\sigma$ is closed and open in $\tilde{X}$. For our purpose, it suffices to work over underlying classical stacks. Consider $X^\square := \text{Spf } R$ and the associated framed map

$$p^\square: \tilde{X}^\square := X^\square \times_X \tilde{X} \longrightarrow X^\square.$$

We need to show that $p^\square$ is an isomorphism towards $X_{\text{Asc}(\sigma)}^\square := \text{Spf } R/(v_i)_{i \notin \text{Asc}(\sigma)}$ over each component labelled by σ .

Indeed, note that the v_i -coordinates in R correspond to the nilpotent matrices $N = \sum_{i=1}^{n-1} v_i E_{i,i+1}$ in the Vogan stack. For each N , its corresponding point lies in $\tilde{X}_\sigma$ if and only if $N \in \text{Lie}(\hat{B}_\sigma)$, or equivalently $w_\sigma N w_\sigma^{-1} \in \text{Lie}(\hat{B})$; since $w_\sigma E_{i,i+1} w_\sigma^{-1} = E_{\sigma(i), \sigma(i+1)}$, this holds if and only if $\sigma(i) < \sigma(i+1)$ for each i such that $v_i \neq 0$, and it hence forces $v_i = 0$ for $i \notin \text{Asc}(\sigma)$. It follows that

$$p^\square(\tilde{X}_\sigma^\square) = X_{\text{Asc}(\sigma)}^\square.$$

On the other hand, there is an evident map $s_\sigma^\square: X_{\text{Asc}(\sigma)}^\square \rightarrow \tilde{X}_\sigma^\square$ given by $(x, N) \mapsto (x, N; \hat{B}_\sigma)$, where x and N denote the semisimple and the nilpotent coordinate of a classical point, respectively. Then $p^\square \circ s_\sigma^\square = \text{id}_{X_{\text{Asc}(\sigma)}^\square}$ and $s_\sigma^\square \circ p^\square = \text{id}_{\tilde{X}_\sigma^\square}$. This completes the proof. $\square$

3.2. The first key calculation. Maintain the notation of the previous section, and let $\lambda = \sum m_i \alpha_i$ be a degree zero weight. Slightly abusively, we write $\mathcal{L}_\lambda$ for the Vogan sheaf regarded as a sheaf on X . The main result in this section is the following.

Theorem 3.2.1. *For $\lambda = \sum_{1 \leq i \leq n-1} m_i \alpha_i$ as above, write $I_\lambda := \{i \in [n-1] : m_i \leq 0\}$. Then for each subset $I \subseteq [n-1]$, we have*

$$R\text{Hom}_X(\mathcal{O}_{X_I}, \mathcal{L}_\lambda) \simeq \begin{cases} k[\sum_{i \in I_\lambda^c} (1 - 2m_i)], & I = I_\lambda, \\ 0, & I \neq I_\lambda. \end{cases}$$

To prove this Theorem 3.2.1, we will explicitly construct a $\hat{T}$ -equivariant projective resolution of $\mathcal{O}_{X_1}$ in terms of finite locally free $\mathcal{O}_X$ -modules.

Construction 3.2.2. Fix an arbitrary subset $S \subseteq [n-1]$. Each term of the resolution $\mathcal{P}_\bullet^S$ will be given as follows:

- Define $\mathbb{N}^S := \{\mathbf{d} = (d_i)_{i \in S} : d_i \in \mathbb{N}\}$;
- For each $\mathbf{d} \in \mathbb{N}^S$, write $|\mathbf{d}| := \sum_{i \in S} d_i$ and define the weight

$$\chi_{\mathbf{d}} := \sum_{i \in S} \lceil d_i/2 \rceil \alpha_i;$$

- For $t \geq 0$, construct the t -th term of $\mathcal{P}_\bullet^S$ by

$$\mathcal{P}_t^S := \bigoplus_{\mathbf{d} \in \mathbb{N}^S, |\mathbf{d}|=t} \mathcal{O}_X(\chi_{\mathbf{d}}).$$

The $\mathcal{P}_t^S$ constructed above is a free $\mathcal{O}_X$ -module with a basis denoted by $\{\mathbf{e}_{\mathbf{d}}\}_{|\mathbf{d}|=t}$. When $t = 0$, the condition $|\mathbf{d}| = 0$ forces $d_i = 0$ for all $i \in S$, and thus $\mathcal{P}_0^S = \mathcal{O}_X$.

Next, we construct the differential map $\partial: \mathcal{P}_{t+1}^S \rightarrow \mathcal{P}_t^S$ by giving the image of each $\mathbf{e}_{\mathbf{d}}$ as follows:

$$\partial(\mathbf{e}_{\mathbf{d}}) = \sum_{i \in S: d_i > 0} (-1)^{\varepsilon(i, \mathbf{d})} \vartheta(d_i) \cdot \mathbf{e}_{\mathbf{d} - \mathbf{e}_i}.$$

The notations in this formula are explained below.

- Given variables v_i, u_i in $\mathcal{O}_X$, set $\vartheta_i(r) = v_i$ (resp. $\vartheta_i(r) = u_i$) when r is odd (resp. even).
- Define $\varepsilon(i, \mathbf{d}) := \sum_{j \in S_{<i}} d_j$, where $S_{<i} = S \cap \{1, \dots, i-1\}$; we also declare $\varepsilon(1, \mathbf{d}) = 0$.
- The $\mathbf{e}_i$ is the $|S|$ -tuple with 1 on its i -th coordinate and 0 elsewhere; set $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i} = 0$ if any coordinate of $\mathbf{d} - \mathbf{e}_i$ is negative.

To finish the construction, we check that the differential map $\partial: \mathcal{P}_{t+1}^S \rightarrow \mathcal{P}_t^S$ above satisfies $\partial \circ \partial = 0$. For this, we compute $(\partial \circ \partial)(\mathbf{e}_{\mathbf{d}})$. This is a linear combination of elements $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i - \mathbf{e}_j}$.

- When $i = j$, the coefficient of $\mathbf{e}_{\mathbf{d} - 2\mathbf{e}_i}$ in $(\partial \circ \partial)(\mathbf{e}_{\mathbf{d}})$ is given by

$$(-1)^{\varepsilon(i, \mathbf{d}) + \varepsilon(i, \mathbf{d} - \mathbf{e}_i)} \vartheta_i(d_i) \vartheta_i(d_i - 1) = (-1)^{\varepsilon(i, \mathbf{d}) + \varepsilon(i, \mathbf{d} - \mathbf{e}_i)} u_i v_i = 0,$$

because we have $u_i v_i = 0$ in $\mathcal{O}_X$.

- When $i \neq j$, the coefficient of $\mathbf{e}_{\mathbf{d} - \mathbf{e}_i - \mathbf{e}_j}$ in $(\partial \circ \partial)(\mathbf{e}_{\mathbf{d}})$ is given by

$$(-1)^{\varepsilon(i, \mathbf{d}) + \varepsilon(j, \mathbf{d} - \mathbf{e}_i)} \vartheta_i(d_i) \vartheta_j(d_j) + (-1)^{\varepsilon(j, \mathbf{d}) + \varepsilon(i, \mathbf{d} - \mathbf{e}_j)} \vartheta_j(d_j) \vartheta_i(d_i).$$

But we always have $\varepsilon(j, \mathbf{d} - \mathbf{e}_i) = \varepsilon(j, \mathbf{d}) - \mathbf{1}_{i < j}$, where $\mathbf{1}_{i < j} \in \{0, 1\}$ is the characteristic function to test whether $i < j$ is true or not; similarly, $\varepsilon(i, \mathbf{d} - \mathbf{e}_j) = \varepsilon(i, \mathbf{d}) - \mathbf{1}_{j < i}$. Thus, for arbitrary i and j , the two exponents of -1 above always differ by 1, so this coefficient equals zero.

Example 3.2.3. For $G = \mathrm{GL}_2$, the resolution in Construction 3.2.2 is nontrivial for $S = \{1\}$. In this case, $\mathbf{d} = d_1 \in \mathbb{N} = \mathbb{N}^S$ and $\chi_{d_1} = \lceil d_1/2 \rceil \alpha_1$. So the resolution is 2-periodic, given by

$$\mathcal{P}_t^S = \mathcal{O}_X(\lceil t/2 \rceil \alpha_1).$$

As for the differential map, we compute $\varepsilon(i, \mathbf{d}) = \varepsilon(i, d_1) = 0$, and hence $\partial(\mathbf{e}_d) = \partial(\mathbf{e}_{d_1}) = \vartheta_1(d_1) \cdot \mathbf{e}_{d_1-1}$. Recall that $\vartheta_1(d_1) = v_1$ for $2 \nmid d_1$ and u_1 for $2 \mid d_1$. To conclude, the resolution is

$$\begin{array}{ccccccc} \cdots & \xrightarrow{u_1} & \mathcal{O}_X(2\alpha_1) & \xrightarrow{v_1} & \mathcal{O}_X(\alpha_1) & \xrightarrow{u_1} & \mathcal{O}_X(\alpha_1) \xrightarrow{v_1} \mathcal{O}_X \twoheadrightarrow \mathcal{O}_{X_{\{1\}}} \longrightarrow 0. \\ & & \parallel & & \parallel & & \parallel \\ & & \mathcal{P}_3 & & \mathcal{P}_2 & & \mathcal{P}_1 \\ & & & & & & \parallel \\ & & & & & & \mathcal{P}_0 \end{array}$$

As a remark, this recovers the resolution constructed for the PGL_2 case (which is essentially the same as the GL_2 case) in Bertoloni-Meli's unpublished notes; see [BM23, §2].

In the rest of this subsection, we will prove Proposition 3.2.6 that, for fixed $I \subseteq [n-1]$, the complex $\mathcal{P}_\bullet^{\mathrm{Ic}}$ gives a projective resolution to $\mathcal{O}_{X_I}$. We will observe from the proof of Proposition 3.2.6 that, after forgetting the $\hat{T}$ -weights, such resolution $\mathcal{P}_\bullet^{\mathrm{Ic}}$ essentially comes from a total complex of certain 2-periodic complexes, which are similar to the GL_2 -case stated above.

Before stating Proposition 3.2.6, we need the following two lemmas as preparation.

Lemma 3.2.4. *The complex $\mathcal{P}_\bullet^{\mathrm{S}}$ given in Construction 3.2.2 is $\hat{T}$ -equivariant.*

Proof. Recall that $\hat{T}$ acts on v_i with weight α_i and acts on u_i with weight 0. Fix $\chi \in \mathbb{X}^*(\hat{T})$. Then the following two multiplication maps of $\mathcal{O}_X$ -modules are $\hat{T}$ -equivariant:

$$v_i: \mathcal{O}_X(\chi) \longrightarrow \mathcal{O}_X(\chi - \alpha_i), \quad u_i: \mathcal{O}_X(\chi) \longrightarrow \mathcal{O}_X(\chi).$$

In Construction 3.2.2, when d_i is odd (resp. even), we have $\vartheta_i(d_i) = v_i$ of $\hat{T}$ -weight α_i (resp. $\vartheta_i(d_i) = u_i$ of $\hat{T}$ -weight 0), and thus $v_i: \mathcal{O}_X(\chi_d) \rightarrow \mathcal{O}_X(\chi_d - \alpha_i)$ (resp. $u_i: \mathcal{O}_X(\chi_d) \rightarrow \mathcal{O}_X(\chi_d)$) is $\hat{T}$ -equivariant. But when d_i is odd (resp. even), there must be $\lceil (d_i - 1)/2 \rceil = \lceil d_i/2 \rceil - 1$ (resp. $\lceil (d_i - 1)/2 \rceil = \lceil d_i/2 \rceil$), so the target of multiplication by v_i (resp. multiplication by u_i) equals $\mathcal{O}_X(\chi_d - \alpha_i) = \mathcal{O}_X(\chi_{d-e_i})$ (resp. $\mathcal{O}_X(\chi_d) = \mathcal{O}_X(\chi_{d-e_i})$), which is the same as in $\mathcal{P}_\bullet^{\mathrm{S}}$. $\square$

Besides the $\hat{T}$ -equivariant property above, we also need the following descent result, which is essentially adapted from [Zhu25, 8.2.3 & 10.6.4] into our context.

Lemma 3.2.5. *Keep the same notations as in the proof of Proposition 3.1.2. Given the atlas map $\pi: X^\square = \mathrm{Spf} R \rightarrow X = (\mathrm{Spf} R)/\hat{T}$, the pullback*

$$\pi^*: \mathrm{QCoh}(X) \simeq \mathrm{QCoh}^{\hat{T}}(X^\square) \longrightarrow \mathrm{QCoh}(X^\square)$$

is exact and conservative.

Proof. The map π is the canonical quotient atlas, and thus a $\hat{T}$ -torsor. Since $\hat{T}$ is a torus, π is smooth, hence flat, and it is surjective by construction. Therefore the pullback functor $\pi^*: \mathrm{QCoh}(X) \rightarrow \mathrm{QCoh}(X^\square)$ is exact.

Notice that $X^\square \times_X X^\square \simeq X^\square \times \hat{T}$, so the Čech nerve of $\pi: X^\square \rightarrow X$ is given as $\pi^\bullet := X^{\square, \bullet} \rightarrow X$ where $X^{\square, n} = X^\square \times_X \cdots \times_X X^\square \simeq X^\square \times \hat{T}^n$. Since π is smooth and surjective, it is an fpqc cover,

and QCoh satisfies fpqc descent. Hence²

$$\mathrm{QCoh}(X) \simeq \mathrm{Tot}(\mathrm{QCoh}(X^{\square, \bullet})) := \lim_{[n] \in \Delta} \mathrm{QCoh}(X^{\square, n}),$$

where under this equivalence the functor π^* is identified with evaluation at degree 0 of the totalization, with $X^{\square, 0} = X^{\square}$. It follows immediately that π^* is conservative. $\square$

Proposition 3.2.6. *Fix a subset $I \subseteq [n-1]$. Taking $S = I^c$ in Construction 3.2.2, the chain complex $\mathcal{P}_{\bullet}^{I^c}$ gives a projective resolution to $\mathcal{O}_{X_I}$.*

Proof. Recall from Proposition 3.1.1 that $\mathcal{O}_X = k[[u_1, \dots, u_n]][v_1, \dots, v_{n-1}]/(u_i v_i)_{1 \leq i \leq n-1}$ and $\mathcal{O}_{X_I} = \mathcal{O}_X/(v_i)_{i \in I^c}$. Note that the $\hat{T}$ -equivariant property of $\mathcal{P}_{\bullet}^{I^c}$ was shown in Lemma 3.2.4, and we may forget the $\hat{T}$ -structure from now on. More precisely, let $\mathfrak{h}\mathcal{P}_{\bullet}^{I^c}$ be the image of $\mathcal{P}_{\bullet}^{I^c}$ along $\pi^*: \mathrm{QCoh}(X) \rightarrow \mathrm{QCoh}(X^{\square})$; then we have

$$\mathfrak{h}\mathcal{P}_t^{I^c} = \bigoplus_{d \in \mathbb{N}^{I^c}, |d|=t} \mathcal{O}_X.$$

By the descent result in Lemma 3.2.5, it suffices to verify the that

- (a) $H_i(\mathfrak{h}\mathcal{P}_{\bullet}^{I^c}) = 0$ for $i > 0$, and
- (b) $H_0(\mathfrak{h}\mathcal{P}_{\bullet}^{I^c}) \simeq \mathcal{O}_X/(v_i)_{i \in I^c}$.

Indeed, this $\mathfrak{h}\mathcal{P}_{\bullet}^{I^c}$ can be interpreted in the following way. For each $i \in I^c$, consider the 2-periodic chain complex

$$\mathfrak{h}\mathcal{P}_{\circ}^{\{i\}} := [\dots \xrightarrow{v_i} \mathcal{O}_X \xrightarrow{u_i} \mathcal{O}_X \xrightarrow{v_i} \mathcal{O}_X] \longrightarrow \mathcal{O}_X/(v_i) \longrightarrow 0.$$

Here, the subscript “ $\circ$ ” denotes the homological degree inside $\mathfrak{h}\mathcal{P}_{\circ}^{\{i\}}$ that is descending from the left, and the last $\mathcal{O}_X$ sits in degree zero. Provided this, it is clear that $\mathfrak{h}\mathcal{P}_{\bullet}^{I^c}$ is isomorphic to the total complex of tensor product of $\mathfrak{h}\mathcal{P}_{\circ}^{\{i\}}$ ’s for $i \in I^c$. Thus, we obtain

$$\mathfrak{h}\mathcal{P}_{\bullet}^{I^c} \simeq (\mathrm{Tot} \mathfrak{h}\mathcal{P}_{\circ}^{I^c})_{\bullet} \longrightarrow \bigotimes_{i \in I^c} \mathcal{O}_X/(v_i) = \mathcal{O}_X/(v_i)_{i \in I^c} = \mathcal{O}_{X_I} \longrightarrow 0,$$

where $\mathfrak{h}\mathcal{P}_{\circ}^{I^c} := \bigotimes_{i \in I^c} \mathfrak{h}\mathcal{P}_{\circ}^{\{i\}}$. Note that $\mathfrak{h}\mathcal{P}_{\circ}^{\{i\}}$ is a free complex with all terms isomorphic to $\mathcal{O}_X$.

We then use induction on $|I^c|$ to show the assertion that $(\mathrm{Tot} \mathfrak{h}\mathcal{P}_{\circ}^{I^c})_{\bullet}$ is acyclic outside degree zero with H_0 isomorphic to $\mathcal{O}_X/(v_i)_{i \in I^c}$. For $|I^c| = 0$, the assertion is tautological. As for the case when $I^c \neq \emptyset$, choose an element $i_0 \in I^c$, and assume that (a) and (b) holds for $I^c \setminus \{i_0\}$. View the tensor product $(\mathrm{Tot} \mathfrak{h}\mathcal{P}_{\circ}^{I^c \setminus \{i_0\}})_{\triangleleft} \otimes \mathfrak{h}\mathcal{P}_{\triangleright}^{\{i_0\}}$ as a bi-complex indexed by $(\triangleleft, \triangleright)$, and we have

$$\begin{aligned} (\mathrm{Tot} \mathfrak{h}\mathcal{P}_{\circ}^{I^c})_{\bullet} &= (\mathrm{Tot} \mathfrak{h}\mathcal{P}_{\circ}^{I^c \setminus \{i_0\}} \otimes \mathfrak{h}\mathcal{P}_{\circ}^{\{i_0\}})_{\bullet} \\ &\simeq (\mathrm{Tot} ((\mathrm{Tot} \mathfrak{h}\mathcal{P}_{\circ}^{I^c \setminus \{i_0\}})_{\triangleleft} \otimes \mathfrak{h}\mathcal{P}_{\triangleright}^{\{i_0\}}))_{\bullet}. \end{aligned}$$

²In terms of Zhu’s terminology, using [Zhu25, Example 10.116], one can deduce from some finiteness properties of X that X is ind-very placid (and hence sind-placid) in the sense of Definition 10.157 of *loc. cit.*. Then applying Proposition 10.106 and Corollary 10.88 of *loc. cit.* renders the same categorical equivalence.

by general property of total complexes. Here we always use subscript “ $\bullet$ ” to represent the index for taking homology. For indices $(p, q) \in \mathbb{Z}_{\geq 0}^2$, there is a homological spectral sequence

$$\mathbf{E}_{p,q}^1 = H_p((\text{Tot } \mathfrak{P}_{\circ}^{\text{I}^c \setminus \{i_0\}})_{\bullet} \otimes \mathfrak{P}_q^{\{i_0\}}) \implies H_{p+q}((\text{Tot } \mathfrak{P}_{\circ}^{\text{I}^c})_{\bullet})$$

that degenerates on $\mathbf{E}^2$ -page. To check this, notice that each $\mathfrak{P}_q^{\{i_0\}}$ is a free $\mathcal{O}_X$ -module, and hence flat, which implies $\mathbf{E}_{p,q}^1 \simeq H_p((\text{Tot } \mathfrak{P}_{\circ}^{\text{I}^c \setminus \{i_0\}})_{\bullet}) \otimes \mathfrak{P}_q^{\{i_0\}}$. By induction hypothesis, $H_p((\text{Tot } \mathfrak{P}_{\circ}^{\text{I}^c \setminus \{i_0\}})_{\bullet}) = 0$ for $p > 0$, and thus only $\mathbf{E}_{0,q}^1$ can be nonzero. We further compute with $p = 0$ that

$$\mathbf{E}_{0,q}^2 = H_q(H_0((\text{Tot } \mathfrak{P}_{\circ}^{\text{I}^c \setminus \{i_0\}})_{\triangleleft}) \otimes \mathfrak{P}_{\bullet}^{\{i_0\}}),$$

and all differentials $\mathbf{E}_{p,q}^r \rightarrow \mathbf{E}_{p-r,q+r-1}^r$ are zero as $p = 0$ implies $p - r < 0$. Therefore, $\mathbf{E}_{p,q}^2 \simeq \mathbf{E}_{p,q}^{\infty}$, and for each $t \geq 0$ we deduce from the spectral sequence that

$$\begin{aligned} H_t((\text{Tot } \mathfrak{P}_{\circ}^{\text{I}^c})_{\bullet}) &\simeq H_t(H_0((\text{Tot } \mathfrak{P}_{\circ}^{\text{I}^c \setminus \{i_0\}})_{\triangleleft}) \otimes \mathfrak{P}_{\bullet}^{\{i_0\}}) \\ &\simeq H_t(\mathcal{O}_X / (v_i)_{i \in \text{I}^c \setminus \{i_0\}} \otimes \mathfrak{P}_{\bullet}^{\{i_0\}}) \end{aligned}$$

where we used the induction hypothesis on H_0 to get the second isomorphism. Further, note that in the complex $\mathcal{O}_X / (v_i)_{i \in \text{I}^c \setminus \{i_0\}} \otimes \mathfrak{P}_{\bullet}^{\{i_0\}}$, each term is isomorphic to $\mathcal{O}_X / (v_i)_{i \in \text{I}^c \setminus \{i_0\}}$ with differential maps u_{i_0} (resp. v_{i_0}) towards the terms of odd (resp. even) degrees; since $i_0 \notin \text{I}^c \setminus \{i_0\}$, both u_{i_0} and v_{i_0} are injective, and one easily sees that $\mathcal{O}_X / (v_i)_{i \in \text{I}^c \setminus \{i_0\}} \otimes \mathfrak{P}_{\bullet}^{\{i_0\}}$ is acyclic outside degree 0. It is also easy to see that

$$H_0(\mathcal{O}_X / (v_i)_{i \in \text{I}^c \setminus \{i_0\}} \otimes \mathfrak{P}_{\bullet}^{\{i_0\}}) \simeq \mathcal{O}_X / (v_i)_{i \in \text{I}^c \setminus \{i_0\}} \otimes_{\mathcal{O}_X} \mathcal{O}_X / (v_{i_0}) \simeq \mathcal{O}_X / (v_i)_{i \in \text{I}^c}.$$

This completes the induction and proves (a) and (b) as desired. So we conclude that the chain complex $\mathcal{P}_{\bullet}^{\text{I}^c}$ gives a projective resolution to $\mathcal{O}_{X_1}$. $\square$

In the rest of this subsection, we use the resolution $\mathcal{P}_{\bullet}^{\text{I}^c} \rightarrow \mathcal{O}_{X_1} \rightarrow 0$ to finish the proof of the theorem.

Proof of Theorem 3.2.1. By our setting, fix the weight $\lambda = \sum_{1 \leq i \leq n-1} m_i \alpha_i$ of degree zero. Recall the usual notation that $\text{I}_{\lambda} = \{i \in [n-1] : m_i \leq 0\}$. The goal is to calculate $R\text{Hom}_X(\mathcal{O}_{X_1}, \mathcal{L}_{\lambda})$ for each subset $\text{I} \subseteq [n-1]$. Recall the construction that $\mathcal{L}_{\lambda} = \iota_* \mathcal{O}_{V_{\varphi}}(\lambda)$ for closed immersion $\iota : V_{\varphi} \hookrightarrow X$, where V_{φ} is defined by setting all $u_i = 0$ for $i = 1, \dots, n$. Thus,

$$R\text{Hom}_X(\mathcal{O}_{X_1}, \mathcal{L}_{\lambda}) \simeq R\text{Hom}_{\mathcal{O}_X}(\mathcal{O}_{X_1}, (\mathcal{O}_X / \mathbf{u})(\lambda))^{\hat{T}},$$

where $\mathcal{O}_X / \mathbf{u}$ is given by setting all $u_i = 0$ in $\mathcal{O}_X = k[[u_1, \dots, u_n]][v_1, \dots, v_{n-1}] / (u_i v_i)_{1 \leq i \leq n-1}$. Also recall that $\hat{T}$ acts on u_i trivially and on v_i by simple root α_i .

Applying internal hom to the resolution $\mathcal{P}_\bullet^{\mathcal{I}^c} \rightarrow \mathcal{O}_{X_I} \rightarrow 0$ in Construction 3.2.2, we get a cochain complex, written as

$$\begin{aligned} \mathcal{C}^{\bullet, \mathcal{I}^c} &:= \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{P}_\bullet^{\mathcal{I}^c}, (\mathcal{O}_X/\mathbf{u})(\lambda)) \\ &\simeq \bigoplus_{d \in \mathbb{N}^{\mathcal{I}^c}, |d|=t} \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X(\chi_d), (\mathcal{O}_X/\mathbf{u})(\lambda)) \\ &\simeq \bigoplus_{d \in \mathbb{N}^{\mathcal{I}^c}, |d|=t} (\mathcal{O}_X/\mathbf{u})(\lambda - \chi_d). \end{aligned}$$

Here, recall from Construction 3.2.2 that $\chi_d = \sum_{i \in \mathcal{I}^c} \lceil d_i/2 \rceil \alpha_i$, which depends on $\mathcal{I}^c$. Taking the $\hat{T}$ -invariant part, each differential map in $(\mathcal{C}^{\bullet, \mathcal{I}^c})^{\hat{T}}$ is directly induced from ∂ in $\mathcal{P}_\bullet^{\mathcal{I}^c}$. To further compute the direct summand $(\mathcal{O}_X/\mathbf{u})(\lambda - \chi_d)^{\hat{T}}$, we will use the following lemma.

Lemma. Let $\nu = \sum_{1 \leq i \leq n-1} r_i \alpha_i$ with $r_i \in \mathbb{Z}$. Then we have

$$\mathcal{O}_X(\nu)^{\hat{T}} = \begin{cases} 0, & \text{if } r_j > 0 \text{ for any } j; \\ v_1^{-r_1} \cdots v_{n-1}^{-r_{n-1}} k[[u_n]][[u_j]]_{r_j=0}, & \text{otherwise.} \end{cases}$$

In particular, $(\mathcal{O}_X/\mathbf{u})(\nu)^{\hat{T}}$ is either isomorphic to k or being 0, determined by whether all $r_i \leq 0$ or not.

Proof of Lemma. Since the $\hat{T}$ -weights of v_i and u_i are respectively α_i and 0, for any monomial of the form $v_1^{a_1} \cdots v_{n-1}^{a_{n-1}} \cdot u_1^{b_1} \cdots u_n^{b_n}$ in $\mathcal{O}_X$, after twisting by $\mathcal{O}_X(\nu)$, its $\hat{T}$ -weight equals $\sum_{1 \leq j \leq n-1} (r_j + a_j) \alpha_j$. So the $\hat{T}$ -invariant condition forces $r_j + a_j = 0$, or equivalently $a_j = -r_j \geq 0$. Also, we have conditions $v_j u_j = 0$ in $\mathcal{O}_X$, so we see if $r_j = 0$ (and hence $a_j = 0$), each u_j is free; else if $r_j < 0$ (and hence $a_j > 0$), then no u_j is allowed, meaning that $b_j = 0$ for all j . $\square$

Applying the lemma renders that $(\mathcal{O}_X/\mathbf{u})(\lambda - \chi_d)^{\hat{T}} \neq 0$ if and only if $\lambda - \chi_d \leq 0$; equivalently, this means $m_i \leq 0$ for each $i \in \mathcal{I}$ and $\lceil d_i/2 \rceil \geq m_i$ for each $i \in \mathcal{I}^c$. In particular, if there is $i \in \mathcal{I}$ with $m_i > 0$, then $(\mathcal{O}_X/\mathbf{u})(\lambda - \chi_d)^{\hat{T}} = 0$ for all $\mathbf{d}$'s, and hence $(\mathcal{C}^{\bullet, \mathcal{I}^c})^{\hat{T}} = 0$; in this case, we clearly have $R\operatorname{Hom}_X(\mathcal{O}_{X_I}, \mathcal{L}_\lambda) = 0$, which holds for all $\mathcal{I} \not\subseteq \mathcal{I}_\lambda$.

From now on, we may assume $\mathcal{I} \subseteq \mathcal{I}_\lambda$, and hence $m_i \leq 0$ for all $i \in \mathcal{I}$. Again, the lemma implies that $(\mathcal{O}_X/\mathbf{u})(\lambda - \chi_d)^{\hat{T}} \simeq k$ and we can choose

$$\mathbf{f}_d := \prod_{j \in \mathcal{I}} v_j^{-m_j} \prod_{i \in \mathcal{I}^c} v_i^{\lceil d_i/2 \rceil - m_i} \in (\mathcal{O}_X/\mathbf{u})(\lambda - \chi_d)^{\hat{T}}$$

as a polynomial generator. Then for varying $\mathbf{d} \in \mathbb{N}^{\mathcal{I}^c}$ with $\lceil d_i/2 \rceil \geq m_i$, these $\mathbf{f}_d$'s form a k -basis of $(\mathcal{C}^{\bullet, \mathcal{I}^c})^{\hat{T}}$. We then describe the differential maps in $(\mathcal{C}^{\bullet, \mathcal{I}^c})^{\hat{T}}$ induced from those in $\mathcal{P}_\bullet^{\mathcal{I}^c}$ over this k -basis. Write $\mathbf{e}_i$ for the $|\mathcal{I}^c|$ -tuple with 1 on its i -th coordinate and 0 elsewhere; according to Construction 3.2.2, for each $i \in \mathcal{I}^c$, the differential map from the summand with index $\mathbf{d}$ towards that with index $\mathbf{d} + \mathbf{e}_i$ is exactly $\vartheta(d_i + 1)$, which is $v_i \neq 0$ (resp. $u_i = 0$) if d_i is even (resp. odd).

Thus, over the basis $\{f_d\}_d$, the differential map in $(\mathcal{C}^{\bullet, I^c})^{\hat{T}}$ is read as

$$f_d \mapsto \sum_{i \in I^c, 2|d_i} (-1)^{\varepsilon(i, d)} f_{d+e_i}.$$

Following the similar idea of proving Proposition 3.2.6, we aim to reinterpret $(\mathcal{C}^{\bullet, I^c})^{\hat{T}}$ as certain total complex of some $C_{\{i\}}^{\circ}$'s, where “ $\circ$ ” denotes the degree index. For each $i \in I^c$, let $C_{\{i\}}^{\circ}$ be the cochain complex whose degrees are bounded from below by $\xi_i := \max\{0, 2m_i - 1\} \in \mathbb{Z}_{\geq 0}$, in which the nonzero terms are 1-dimensional k -vector spaces $C_{\{i\}}^r = k \cdot g_{i,r}$, where $g_{i,r}$ are the generators with $r \geq \max\{0, 2m_i - 1\}$. The differential maps in $C_{\{i\}}^{\circ}$, denoted by d_C^r , are taken as

$$g_{i,r} \mapsto \begin{cases} g_{i,r+1}, & \text{if } r \text{ even,} \\ 0, & \text{if } r \text{ odd.} \end{cases}$$

In other words,

$$C_{\{i\}}^{\circ} = [0 \xrightarrow{d_C^0} \cdots \longrightarrow 0 \longrightarrow k \cdot g_{i,\xi_i} \xrightarrow{d_C^{\xi_i}} k \cdot g_{i,\xi_i+1} \longrightarrow \cdots].$$

We observe that $\lceil d_i/2 \rceil \geq m_i$ is equivalent to $d_i \geq \xi_i = \max\{0, 2m_i - 1\}$, and that ξ_i is odd if and only if $m_i > 0$. Thus, by construction of $C_{\{i\}}^{\circ}$, we can choose an isomorphism

$$(\mathcal{C}^{\bullet, I^c})^{\hat{T}} \simeq (\text{Tot } C_{I^c})^{\bullet}$$

where we write $C_{I^c} := \bigotimes_{i \in I^c} C_{\{i\}}^{\circ}$ for simplicity. The argument above works without assuming $I \subseteq I_{\lambda}$.

Now it still remains to compute cohomology of each $C_{\{i\}}^{\circ}$, but this is clear from our construction. We note that $\xi_i = 0$ if and only if $m_i \leq 0$, and then obtain

$$H^r(C_{\{i\}}^{\circ}) \simeq \begin{cases} k, & \text{if } r = 2m_i - 1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, the cohomology of $(\text{Tot } C_{I^c})^{\bullet}$ is nonzero if and only if $m_i > 0$ for all $i \in I^c$, or equivalently $I = I_{\lambda}$.

To conclude, for $I \neq I_{\lambda}$ the cohomology of $(\mathcal{C}^{\bullet, I^c})^{\hat{T}}$ always vanishes, and so also does the desired $R\text{Hom}$. When $I = I_{\lambda}$, there is a quasi-isomorphism $C_{\{i\}}^{\circ} \approx k[-(2m_i - 1)]$ between complexes of k -vector spaces for each $i \in I^c = I_{\lambda}^c$; it then deduces a quasi-isomorphism $(\text{Tot } C_{I_{\lambda}^c})^{\bullet} \approx k[\sum_{i \in I_{\lambda}^c} (1 - 2m_i)]$, where the right side is a derived complex concentrated in a single degree. This finishes the proof of Theorem 3.2.1. $\square$

3.3. Maps between Vogan sheaves. In this section, we completely determine the hom space $R\text{Hom}(\mathcal{L}_{\lambda}, \mathcal{L}_{\mu})$ for any weights λ, μ . It is clear that this hom vanishes identically if λ and μ are of unequal degrees. If they are of equal degrees, we may uniquely write $\mu = \lambda + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ as usual.

Theorem 3.3.1. *In the notation above, with $I_{\mu-\lambda} = \{i \in [n-1] : m_i \leq 0\}$, we have*

$$R\text{Hom}(\mathcal{L}_{\lambda}, \mathcal{L}_{\mu}) = k[-\sum_{i \in I_{\mu-\lambda}^c} 2m_i] \oplus k[-1 - \sum_{i \in I_{\mu-\lambda}^c} 2m_i].$$

We emphasize that the odd degree class here is “junk”, and would disappear if we were working with PGL_n instead of GL_n . However, the unique even degree class is of basic importance. Note in particular that when $\mu = \lambda \pm \alpha$ for some simple root $\alpha \in \mathbb{X}^*(\hat{T})$, the even degree class lives in degree two and corresponds to a canonical map

$$\begin{aligned} L_\alpha: \mathcal{L}_\lambda &\longrightarrow \mathcal{L}_{\lambda-\alpha}, \\ \text{resp. } R_\alpha: \mathcal{L}_\lambda &\longrightarrow \mathcal{L}_{\lambda+\alpha}[2], \end{aligned}$$

according to the sign. As in the introduction, we call these lowering and raising operators. The proof below will show that these generate all the Homs between Vogan sheaves, in the sense that the even degree class prescribed by the theorem is uniquely a composition of lowering operators (in the indices with $m_i < 0$) and raising operators (in the indices with $m_i > 0$).

Proof of Theorem 3.3.1. We split the argument into the following several steps.

Step I (Computing the derived self-intersection). As $\mathcal{O}_{V_\varphi}(\lambda)$ is an invertible weight line bundle on V_φ , we directly compute

$$\begin{aligned} R\mathrm{Hom}(\mathcal{L}_\lambda, \mathcal{L}_\mu) &\simeq R\mathrm{Hom}(\iota_* \mathcal{O}_{V_\varphi}(\lambda), \iota_* \mathcal{O}_{V_\varphi}(\mu)) \\ &\simeq R\mathrm{Hom}_{V_\varphi}(\iota^* \iota_* \mathcal{O}_{V_\varphi}, \mathcal{O}_{V_\varphi}(\mu - \lambda)). \end{aligned}$$

This reduces to the case of degree-zero weights. We then rewrite $\iota^* \iota_* \mathcal{O}_{V_\varphi}$ in terms of self-intersection of V_φ in Par_G , i.e., note that $\iota^* \iota_* \mathcal{O}_{V_\varphi} \simeq \mathcal{O}_{V_\varphi \times_{\mathrm{Par}_G} V_\varphi} \simeq \iota_* \mathcal{O}_{V_\varphi} \otimes_{\mathbb{L} \mathcal{O}_{\mathrm{Par}_G}} \iota_* \mathcal{O}_{V_\varphi}$. Since X is the formal completion of Par_G with respect to V_φ , this tensor product can be viewed as over $\mathcal{O}_X$, and thus

$$\iota^* \iota_* \mathcal{O}_{V_\varphi} \simeq \iota_* \mathcal{O}_{V_\varphi} \otimes_{\mathbb{L} \mathcal{O}_X} \iota_* \mathcal{O}_{V_\varphi}.$$

To further compute this, we need the following algebraic fact.

Lemma. Given a ring $R = R_0[[t]]$, for any R_0 -modules M and N viewed as R -modules via the natural quotient map $R \rightarrow R_0$, there is a natural isomorphism

$$M \otimes_R^{\mathbb{L}} N \simeq (M \otimes_{R_0}^{\mathbb{L}} N) \oplus (M \otimes_{R_0}^{\mathbb{L}} N)[1].$$

Proof of Lemma. Since t is a nonzero divisor on R , the exact sequence $0 \rightarrow R \xrightarrow{t} R \rightarrow R_0 \rightarrow 0$ gives a free R -resolution of R_0 in two consecutive degrees $[-1, 0]$. It follows that

$$M \otimes_R^{\mathbb{L}} R_0 \simeq M \otimes_R [R \xrightarrow{t} R] = [M \xrightarrow{0} M] = M \oplus M[1].$$

Applying the derived base change formula [Sta18, 06Y5], we see

$$M \otimes_R^{\mathbb{L}} N \simeq (M \otimes_R^{\mathbb{L}} R_0) \otimes_{R_0}^{\mathbb{L}} N \simeq (M \oplus M[1]) \otimes_{R_0}^{\mathbb{L}} N.$$

This gives the desired right side of the lemma. $\square$

Now apply the lemma to $R = \mathcal{O}_X$, $t = u_n$, and $M = N = \iota_* \mathcal{O}_{V_\varphi}$. Notice that $\iota_* \mathcal{O}_{V_\varphi} = \mathcal{O}_X / \mathbf{u} = k[v_1, \dots, v_{n-1}]$ by setting all $u_i = 0$ in $\mathcal{O}_X$. To simplify the notation, we write

$$\Upsilon := \iota_* \mathcal{O}_{V_\varphi} \otimes_{\mathbb{L} \mathcal{O}_X / (u_n)} \iota_* \mathcal{O}_{V_\varphi}.$$

This is the sheaf of $\mathcal{O}_X/(u_n)$ -module for derived self-intersection of V_φ . Then the lemma above deduces a natural isomorphism

$$\iota^* \iota_* \mathcal{O}_{V_\varphi} \simeq \Upsilon \oplus \Upsilon[1],$$

and it further follows that

$$\begin{aligned} R\mathrm{Hom}(\mathcal{L}_\lambda, \mathcal{L}_\mu) &\simeq R\mathrm{Hom}_{V_\varphi}(\Upsilon, \mathcal{O}_{V_\varphi}(\mu - \lambda)) \oplus R\mathrm{Hom}_{V_\varphi}(\Upsilon[1], \mathcal{O}_{V_\varphi}(\mu - \lambda)) \\ &\simeq R\mathrm{Hom}_{V_\varphi}(\Upsilon, \mathcal{O}_{V_\varphi}(\mu - \lambda)) \oplus R\mathrm{Hom}_{V_\varphi}(\Upsilon, \mathcal{O}_{V_\varphi}(\mu - \lambda))[-1]. \end{aligned}$$

Thus, to complete proving Theorem 3.3.1, it suffices to show that

$$R\mathrm{Hom}_{V_\varphi}(\Upsilon, \mathcal{O}_{V_\varphi}(\mu - \lambda)) = k[-\sum_{i \in I_{\mu-\lambda}^c} 2m_i].$$

This simplifies the argument into a single-degree case.

Step II (Construction of finite Koszul-type resolution). For this purpose, with the notations in the lemma above, write $R_0 = \mathcal{O}_X/(u_n)$ and hence $\iota_* \mathcal{O}_{V_\varphi} \simeq R_0/(u_1, \dots, u_{n-1})$ as R_0 -modules. Using the idea similar to Construction 3.2.2, we will construct the following finite complex $\mathcal{K}^\bullet$, in terms of R_0 -modules, and then show that the natural augmentation $\mathcal{K}^\bullet \rightarrow \iota_* \mathcal{O}_{V_\varphi}$ is a quasi-isomorphism.

Since $u_i v_i = 0$ for $i = 1, \dots, n-1$ in R_0 , we obtain a short exact sequence

$$0 \longrightarrow R_0/(v_i) \xrightarrow{u_i} R_0 \longrightarrow R_0/(u_i) \longrightarrow 0.$$

This gives a resolution $K_{\{i\}}^\star := [R_0/(v_i) \xrightarrow{u_i} R_0]$ of R_0 -modules to $R_0/(u_i)$ in degrees $[-1, 0]$, indexed with $\star$. Then consider the total complex

$$\mathcal{K}^\bullet := \mathrm{Tot}(K_{\{1\}}^\star \otimes_{R_0} \cdots \otimes_{R_0} K_{\{n-1\}}^\star)$$

in which, for each index $0 \leq p \leq n-1$, the p -th graded piece is given by

$$\mathcal{K}^{-p} = \bigoplus_{\substack{S \subseteq [n-1], \\ |S|=p}} R_0/(v_i)_{i \in S} = \bigoplus_{\substack{S \subseteq [n-1], \\ |S^c|=p}} \mathcal{O}_{X_S}.$$

On the graded pieces, the differential maps $\mathcal{K}^{-p} \rightarrow \mathcal{K}^{-p+1}$ of $\mathcal{K}^\bullet$ are described as follows. For subset $S \subseteq [n-1]$ with $|S| = p$, define $\varepsilon(i, S) := |S_{<i}| = |S \cap \{1, \dots, i-1\}|$. Then on each direct summand $R_0/(v_i)_{i \in S} \subseteq \mathcal{K}^{-p}$, the differential map $\mathcal{K}^{-p} \rightarrow \mathcal{K}^{-p+1}$ behaves as $x \mapsto \sum_{i \in S} (-1)^{\varepsilon(i, S)} u_i x$, with image landing in $\sum_{i \in S} R_0/(v_i)_{i \in S \setminus \{i\}}$.

To further simplify the notations, write $K_S := \bigotimes_{i \in S} K_{\{i\}}^\star$ for each subset $S \subseteq [n-1]$. It then suffices to check that

$$(\mathrm{Tot} K_S)^\bullet \longrightarrow R_0/(u_i)_{i \in S}$$

is a quasi-isomorphism for each S , as taking $S = [n-1]$ in this assertion gives that

$$\mathcal{K}^\bullet = (\mathrm{Tot} K_{[n-1]})^\bullet \xrightarrow{\sim} R_0/(u_1, \dots, u_{n-1}) \simeq \iota_* \mathcal{O}_{V_\varphi}$$

is a quasi-isomorphism. We argue by induction on $|S|$. The case $S = \emptyset$ is tautological. Now let $S \neq \emptyset$ and choose $s_0 \in S$. Assume the claim holds for $S \setminus \{s_0\}$. We note that

$$(\mathrm{Tot} K_S)^\bullet \simeq (\mathrm{Tot} ((\mathrm{Tot} K_{S \setminus \{s_0\}})^\triangleleft \otimes_{R_0} K_{\{s_0\}}^\triangleright))^\bullet$$

where the right side is the total complex of a bounded bi-complex indexed by $(\triangleleft, \triangleright)$, with $-|S| + 1 \leq \triangleleft \leq 0$ and $-1 \leq \triangleright \leq 0$. Filtering the right side above by the second grading yields a finite spectral sequence

$$\mathbf{E}_1^{p,q} = H^q((\text{Tot } K_{S \setminus \{s_0\}})^p \otimes_{R_0} K_{\{s_0\}}^\bullet) \implies H^{p+q}((\text{Tot } K_S)^\bullet).$$

Here we always use “ $\bullet$ ” to indicate the cohomological index. By construction, each term $(\text{Tot } K_{S \setminus \{s_0\}})^p$ is a finite direct sum of R_0 -modules of the form $R_0/(v_i)_{i \in T}$ for some $T \subseteq S \setminus \{s_0\}$ with $|T| = -p$; for such subset T , the corresponding summand of the complex $(\text{Tot } K_{S \setminus \{s_0\}})^p \otimes_{R_0} K_{\{s_0\}}^\bullet$ is

$$[R_0/(v_i)_{i \in T \cup \{s_0\}} \xrightarrow{u_{s_0}} R_0/(v_i)_{i \in T}].$$

Notice that u_{s_0} is a non-zero-divisor in $R_0/(v_i)_{i \in T \cup \{s_0\}}$, and hence multiplication by u_{s_0} is injective. Then for the complex above in degrees $[-1, 0]$, we have $H^{-1} = 0$ and $H^0 \simeq R_0/(u_{s_0}, (v_i)_{i \in T})$. It follows that

(a) $\mathbf{E}_1^{p,-1} = 0$ for all $-|S| + 1 \leq p \leq 0$;

(b) $\mathbf{E}_1^{p,0}$ is identified with the degree- p term of $(\text{Tot } \bigotimes_{i \in S \setminus \{s_0\}} [R_0/(u_{s_0}, v_i) \xrightarrow{u_{s_0}} R_0/(u_{s_0})])^\bullet$.

By the induction hypothesis, the total complex in (b) is quasi-isomorphic to $R_0/(u_{s_0}, (u_i)_{i \in S \setminus \{s_0\}}) = R_0/(u_i)_{i \in S}$. Hence $\mathbf{E}_2^{p,q} = 0$ unless $(p, q) = (0, 0)$, while $\mathbf{E}_2^{0,0} \simeq R_0/(u_i)_{i \in S}$. Therefore, we conclude that $\mathbf{E}_2^{p,q} \simeq \mathbf{E}_\infty^{p,q}$ and

$$H^r((\text{Tot } K_S)^\bullet) \simeq \begin{cases} R_0/(u_i)_{i \in S}, & r = 0, \\ 0, & r \neq 0. \end{cases}$$

The induced map on $H^0((\text{Tot } K_S)^\bullet)$ is the natural quotient map, hence an isomorphism. To conclude, we obtain a quasi-isomorphism $(\text{Tot } K_S)^\bullet \approx R_0/(u_i)_{i \in S}$; in particular, there is a quasi-isomorphism

$$\mathcal{K}^\bullet \approx \iota_* \mathcal{O}_{V_\varphi}.$$

where $\iota_* \mathcal{O}_{V_\varphi}$ is viewed as a complex concentrated in a single degree.

Step III (Finishing the proof). The quasi-isomorphism above then implies that

$$\begin{aligned} R\text{Hom}_{V_\varphi}(\Upsilon, \mathcal{O}_{V_\varphi}(\mu - \lambda)) &\simeq R\text{Hom}_{V_\varphi}(\iota^* \mathcal{K}^\bullet, \mathcal{O}_{V_\varphi}(\mu - \lambda)) \\ &\simeq R\text{Hom}_X(\mathcal{K}^\bullet, \iota_* \mathcal{O}_{V_\varphi}(\mu - \lambda)) \\ &= R\text{Hom}_X(\mathcal{K}^\bullet, \mathcal{L}_{\mu - \lambda}). \end{aligned}$$

For our purpose, we need to identify the derived complex $R\text{Hom}_X(\mathcal{K}^\bullet, \mathcal{L}_{\mu - \lambda})$ with $k[-\sum_{i \in I_{\mu - \lambda}^c} 2m_i]$ that is concentrated in a single degree. Following a standard spectral sequence argument as in the proof of Proposition 3.2.6, this requires to compute for each index $0 \leq p \leq n - 1$ that

$$\bigoplus_{\substack{S \subseteq [n-1], \\ |S^c| = p}} R\text{Hom}_X(\mathcal{O}_{X_S}, \mathcal{L}_{\mu - \lambda})[-p].$$

Notice that $\mu - \lambda$ is of degree 0, and thus Theorem 3.2.1 applies, from which we obtain that

$$R\text{Hom}_X(\mathcal{O}_{X_S}, \mathcal{L}_{\mu - \lambda}) = 0, \quad \text{unless } S = I_{\mu - \lambda}.$$

Thus, the only nonzero term is at $p = |\mathbb{I}_{\mu-\lambda}^c|$. It follows that

$$\begin{aligned} R\mathrm{Hom}_{V_\varphi}(\Upsilon, \mathcal{O}_{V_\varphi}(\mu - \lambda)) &\simeq R\mathrm{Hom}_X(\mathcal{O}_{X_{\mathbb{I}_{\mu-\lambda}}}, \mathcal{L}_{\mu-\lambda})[-|\mathbb{I}_{\mu-\lambda}^c|] \\ &\simeq k[\sum_{i \in \mathbb{I}_{\mu-\lambda}^c} (1 - 2m_i) - |\mathbb{I}_{\mu-\lambda}^c|] \\ &= k[-\sum_{i \in \mathbb{I}_{\mu-\lambda}^c} 2m_i], \end{aligned}$$

where the second isomorphism is also by Theorem 3.2.1. This completes the proof of Theorem 3.3.1. $\square$

4. STALKS AT BASIC POINTS I: SPLIT CASE

In this section, using Theorem 3.2.1, we prove Theorem 1.2.1 for weights of degree zero, that is, we compute the stalk of $\mathcal{L}_\lambda^{\mathrm{aut}}$ at $b = 1$.

4.1. Proof of Theorem 1.2.1 for degree zero. Fix a degree zero weight $\lambda = \sum_{1 \leq i \leq n-1} m_i \alpha_i$, and set $\mathbb{I} = \mathbb{I}_\lambda = \{i \in [n-1] : m_i \leq 0\}$ as usual. We deduce Theorem 1.2.1 in this case of degree zero, and the proof of the general case will be in Subsection 5.3.

By the general reasoning in the previous section, we know that $i_1^* \mathcal{L}_\lambda^{\mathrm{aut}}$ is of finite length, and is built entirely from the representations $\pi_{\mathbb{I}}$ (for varying $\mathbb{I}$) by iterated shifts and extensions. Moreover, we know that the Jacquet module functor r_G^B is exact and conservative on the Iwahori block, and it separates the $\pi_{\mathbb{I}}$'s in the sense that $r_G^B \pi_{\mathbb{I}}$ and $r_G^B \pi_{\mathbb{J}}$ do not share any common subquotient whenever $\mathbb{I} \neq \mathbb{J}$. Therefore, it suffices to compute $r_G^B i_1^* \mathcal{L}_\lambda^{\mathrm{aut}}$ and match this with the Jacquet module of the right side of Theorem 1.2.1.

Now, the expression $r_G^B i_1^* \mathcal{L}_\lambda^{\mathrm{aut}}$ can be rewritten as $i_1^* \mathrm{CT}_{B!} \mathcal{L}_\lambda^{\mathrm{aut}}$, using the results in [HHS24]. Since the Langlands functor $\mathbf{L}_G$ is an equivalence and is compatible with Eisenstein series, it is also compatible with constant terms. In particular, $\mathrm{CT}_{B!} \mathcal{L}_\lambda^{\mathrm{aut}}$ matches the sheaf $\mathrm{CT}_B^{\mathrm{spec}} \mathcal{L}_\lambda$ under categorical local Langlands for the torus T . It is clear that $\mathrm{CT}_B^{\mathrm{spec}} \mathcal{L}_\lambda$ is set-theoretically supported on the orbit $\{\sigma\varphi\} \subseteq |\mathrm{Par}_T^{\mathrm{unr}}| = \hat{T}$, so this sheaf inherits a canonical direct sum decomposition by the points in this orbit, which we write as

$$\mathrm{CT}_B^{\mathrm{spec}} \mathcal{L}_\lambda = \bigoplus_{\sigma \in S_n} (\mathrm{CT}_B^{\mathrm{spec}} \mathcal{L}_\lambda)^\sigma.$$

Moreover, under categorical Langlands for tori, restricting to the point $b = 1$ corresponds to passing to the weight zero summand of the associated coherent sheaf, which on $\mathrm{Par}_T^{\mathrm{unr}} = \hat{T} \times \mathbb{B}\hat{T}$ is equivalent to taking global sections. In summary, this argument shows the following result.

Proposition 4.1.1. *For any $i \in \mathbb{Z}$ and any $\sigma \in S_n$, the multiplicity of $\sigma\delta^{1/2}$ in $H^i(r_G^B i_1^* \mathcal{L}_\lambda^{\mathrm{aut}})$ equals the dimension of $H^i(R\Gamma(\mathrm{Par}_T, (\mathrm{CT}_B^{\mathrm{spec}} \mathcal{L}_\lambda)^\sigma))$.*

Comparing this with the expected answer via Proposition 2.1.2 (1), we have reduced our goal to proving that $R\Gamma(\mathrm{Par}_T, (\mathrm{CT}_B^{\mathrm{spec}} \mathcal{L}_\lambda)^\sigma)$ vanishes unless $\mathrm{Asc}(\sigma) = \mathbb{I}$, in which case it is one-dimensional and concentrated in the expected degree.

We now analyze these global sections further. Since $\mathrm{CT}_B^{\mathrm{spec}} = \mathbf{q}_* \mathbf{p}^!$ along $\mathrm{Par}_T \xleftarrow{\mathbf{q}} \mathrm{Par}_B \xrightarrow{\mathbf{p}} \mathrm{Par}_G$, it is clear that

$$R\Gamma(\mathrm{Par}_T, \mathrm{CT}_B^{\mathrm{spec}} \mathcal{L}_\lambda) \simeq R\Gamma(\mathrm{Par}_B, \mathbf{p}^! \mathcal{L}_\lambda).$$

Now $\mathfrak{p}^! \mathcal{L}_\lambda$ is supported on the preimage of V_φ along $\mathfrak{p}$, so we can equally well write this as $R\Gamma(\tilde{X}, \mathfrak{p}^! \mathcal{L}_\lambda)$ where we abusively view $\mathcal{L}_\lambda$ as a sheaf on X ; recall that $p: \tilde{X} \rightarrow X$ with $\tilde{X}$ the formal completion of Par_B as in the previous section. It is then formal that the decomposition of $\tilde{X}$ into the connected components $\tilde{X}_\sigma$ induces the decomposition of these global sections into their S_n -summands. In other words, the isomorphism

$$R\Gamma(\text{Par}_T, \text{CT}_B^{\text{spec}} \mathcal{L}_\lambda) \simeq R\Gamma(\tilde{X}, \mathfrak{p}^! \mathcal{L}_\lambda)$$

obtained in the above analysis restricts to compatible isomorphisms

$$R\Gamma(\text{Par}_T, (\text{CT}_B^{\text{spec}} \mathcal{L}_\lambda)^\sigma) \simeq R\Gamma(\tilde{X}_\sigma, \mathfrak{p}_\sigma^! \mathcal{L}_\lambda).$$

This is an immediate consequence of the fact that the reduced subscheme of $\tilde{X}_\sigma$ maps via q onto the point $\sigma\varphi$. Since p_σ is a closed immersion and factors over an isomorphism $\tilde{X}_\sigma \simeq X_{\text{Asc}(\sigma)}$, we can rewrite the right side as

$$R\text{Hom}(p_{\sigma*} \mathcal{O}_{\tilde{X}_\sigma}, \mathcal{L}_\lambda) \simeq R\text{Hom}_X(\mathcal{O}_{X_{\text{Asc}(\sigma)}}, \mathcal{L}_\lambda)$$

using that $p_{\sigma*}$ is left adjoint to $\mathfrak{p}_\sigma^!$. In summary, we have proved that

$$R\Gamma(\text{Par}_T, (\text{CT}_B^{\text{spec}} \mathcal{L}_\lambda)^\sigma) \simeq R\text{Hom}_X(\mathcal{O}_{X_{\text{Asc}(\sigma)}}, \mathcal{L}_\lambda).$$

The result now follows from Theorem 3.2.1, which completely describes the rightmost hom space appearing here.

4.2. Variant for Isoc. We briefly explain how a version of the same argument applies unconditionally in Zhu's setting. More precisely, the sheaf $\mathcal{L}_\lambda$ is supported on the unipotent part of the parameter stack denoted by $\text{Loc}_c^{\text{unip}} G$ in [Zhu25], so Zhu's equivalence gives a sheaf $\mathcal{L}_\lambda^{\text{Isoc}} \in \text{Shv}(\text{Isoc}_G)$.

Theorem 4.2.1. *For any degree zero weight $\lambda = \sum m_i \alpha_i$, we have*

$$i_1^! \mathcal{L}_\lambda^{\text{Isoc}} = \pi_I[\sum_{i \in I^c} (1 - 2m_i)]$$

where $I = \{i \in [n-1] : m_i \leq 0\}$.

Proof. Let $\Pi = \mathcal{C}_c(G(E)/I) = i_{B-}^G \mathcal{C}_c(T(E)/T(\mathcal{O}_E))$ be the Bernstein generator of the Iwahori block. Under Zhu's equivalence in *loc. cit.*, the sheaf $i_{1!} \Pi$ matches the coherent Springer sheaf $\mathcal{S} = \mathfrak{p}_* \mathcal{O}_{\text{Par}_B^{\text{unr}}}$. Moreover, by [Yan25, Lemma 3.14] and [YZ25, Proposition 4.43] this equivalence matches the canonical injective map $\mathcal{O}(\hat{T}) \rightarrow \text{End}(\mathcal{S})$ induced by the identification $\mathcal{O}(\hat{T}) = \mathcal{O}(\text{Par}_B^{\text{unr}})$ with the canonical copy of $\mathcal{O}(\hat{T}) = \mathcal{C}_c(T(E)/T(\mathcal{O}_E))$ inside $\text{End}(\Pi)$. Using these facts we compute

$$\begin{aligned} R\text{Hom}_{\text{Rep}}(\mathcal{C}_c(T(E)/T(\mathcal{O}_E)), r_G^B i_1^! \mathcal{L}_\lambda^{\text{Isoc}}) &\simeq R\text{Hom}_{\text{Rep}}(\Pi, i_1^! \mathcal{L}_\lambda^{\text{Isoc}}) \\ &\simeq R\text{Hom}_{\text{Isoc}}(i_{1!} \Pi, \mathcal{L}_\lambda^{\text{Isoc}}) \\ &\simeq R\text{Hom}_{\text{Loc}}(\mathcal{S}, \mathcal{L}_\lambda), \end{aligned}$$

where in the first line we used definition of Π along with second adjointness, and in the last line we used Zhu's equivalence. This string of isomorphisms is compatible with the natural actions of $\mathcal{O}(\hat{T})$ as discussed above.

Now, since $\mathcal{C}_c(T(E)/T(\mathcal{O}_E))$ is a minimal projective generator for the block of unramified $T(E)$ -representations, it is clear from the above calculation that the multiplicity of $\sigma \delta^{1/2}$ in $H^i(r_G^B i_1^! \mathcal{L}_\lambda^{\text{Isoc}})$

agrees with the length of $H^i(R\mathrm{Hom}(\mathcal{S}, \mathcal{L}_\lambda)_\sigma)$, where the subscript indicates localization at the maximal ideal cutting out $\sigma\varphi$ in $\mathcal{O}(\hat{T})$. On the other hand, arguing as in the previous proof we know that

$$\begin{aligned} R\mathrm{Hom}(\mathcal{S}, \mathcal{L}_\lambda) &= R\mathrm{Hom}_X(p_*\mathcal{O}_{\tilde{X}}, \mathcal{L}_\lambda) \\ &= \bigoplus_{\sigma \in S_n} R\mathrm{Hom}_X(p_{\sigma*}\mathcal{O}_{\tilde{X}_\sigma}, \mathcal{L}_\lambda), \end{aligned}$$

and it is easy to see that the σ -indexed summand in the last line here is isomorphic to $R\mathrm{Hom}(\mathcal{S}, \mathcal{L}_\lambda)_\sigma$. Putting all this together, we see that the multiplicity of $\sigma\delta^{1/2}$ in $H^i(r_G^B i_1^! \mathcal{L}_\lambda^{\mathrm{Isoc}})$ agrees with the dimension of $H^i(R\mathrm{Hom}_X(p_{\sigma*}\mathcal{O}_{\tilde{X}_\sigma}, \mathcal{L}_\lambda))$, and from here one concludes exactly as in the previous proof. $\square$

5. STALKS AT BASIC POINTS II: GENERAL CASE

In this section we compute the stalks of $\mathcal{L}_\lambda^{\mathrm{aut}}$ at all basic points. Throughout this section we assume that c_ψ is compatible with Eisenstein series, and freely appeal to the consequences of this assumption as discussed in Section 2.2.

5.1. Anisotropic points. In this subsection we assume that $(d, n) = 1$, so G_{b_d} is compact mod center. In this case, the trivial representation $\mathbf{1}$ is the only irreducible representation of $G_{b_d}(E)$ with L-parameter φ .

Theorem 5.1.1. *Assume $(d, n) = 1$. Then we have*

$$\mathcal{L}_{\omega_d}^{\mathrm{aut}} = i_{b_d!} \mathbf{1}.$$

Proof. By a fundamental result of Bertolini Meli–Koshikawa, ω_d is a basic weight, so $\mathcal{L}_{\omega_d}^{\mathrm{aut}}$ is of the form $i_{b_d!} \Pi$ for some $\Pi \in D(G_{b_d}(E), k)$ whose cohomology is of finite length. Moreover, the only irreducible representation occurring in Π is $\mathbf{1}$. We also observe that since G_{b_d} is anisotropic mod center, and its center has split rank one, the global dimension of the category of smooth representations of this group is one, so we can choose an isomorphism

$$\Pi \simeq \bigoplus_i H^i(\Pi)[-i].$$

Since Π is the direct sum of its shifted cohomologies, it is clear that the isomorphism above deduces an injection

$$\bigoplus_i \mathrm{End}(H^i(\Pi)) \hookrightarrow \mathrm{End}(\Pi).$$

On the other hand, using the categorical equivalence, we know that

$$\mathrm{End}(\Pi) = \mathrm{End}(\mathcal{L}_{\omega_d}^{\mathrm{aut}}) = \mathrm{End}(\mathcal{L}_{\omega_d}).$$

To proceed further, we apply Theorem 3.3.1 and compute

$$\mathrm{End}(\mathcal{L}_{\omega_d}) = H^0(R\mathrm{Hom}(\mathcal{L}_{\omega_d}, \mathcal{L}_{\omega_d})) = k.$$

Combining these, the one-dimensionality of $\mathrm{End}(\Pi)$ implies that $H^i(\Pi) \neq 0$ for exactly one i . In other words,

$$\Pi = \pi[s]$$

for some $s \in \mathbb{Z}$ and some finite-length smooth π whose only irreducible subquotient is $\mathbf{1}$. It remains to see that $\pi = \mathbf{1}$ and $s = 0$.

By the calculations above, we know that $\text{End}(\pi) = k$. Now, the subcategory $\mathcal{C}$ of finite-length smooth $G_{b_d}(E)$ -representations generated by $\mathbf{1}$ is equivalent to the category of pairs (V, α) where V is a finite-dimensional k -vector space and α is a nilpotent endomorphism, and under this equivalence we have $\text{length}(\pi) = \dim V_\pi$. More precisely, if $K \subseteq G_{b_d}(E)$ is a maximal compact subgroup, the functor $\pi \mapsto \pi^K$ induces an equivalence from the principal block towards the category of $k[T^{\pm 1}]$ -modules, and this restricts to an equivalence from $\mathcal{C}$ towards the category of finite-length $k[T^{\pm 1}]$ -modules V supported at $T = 1$. The latter is equivalent to the category of finite-dimensional vector spaces equipped with a nilpotent endomorphism, by setting $\alpha = T - 1$. Now an easy calculation shows that $\text{End}(V, \alpha) = k$ if and only if $\dim V = 1$, so we see that π has length one, and therefore $\pi = \mathbf{1}$.

It remains to see that $s = 0$. For this we use the duality compatibility of the categorical local Langlands equivalence. More precisely, we know from Proposition 2.3.4 that $\mathbb{D}_{\text{BZ}} \mathcal{L}_{\omega_d}^{\text{aut}} = \mathcal{L}_{\omega_d}^{\text{aut}}[-1]$. On the other hand, by our analysis so far we have $\mathcal{L}_{\omega_d}^{\text{aut}} = i_{b_d!} \mathbf{1}[s]$, so a direct calculation gives

$$\mathbb{D}_{\text{BZ}} i_{b_d!} \mathbf{1}[s] \simeq i_{b_d!} \mathbb{D}_{\text{coh}} \mathbf{1}[s] \simeq i_{b_d!} \mathbf{1}[-1-s],$$

and comparing these results shows that $s = 0$ as desired. $\square$

From this seed case, we deduce the general result at anisotropic points.

Theorem 5.1.2. *For any weight $\lambda = \omega_d + \sum_i m_i \alpha_i$ of degree d , with $(d, n) = 1$, we have*

$$i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}} = \mathbf{1}[-\sum_{i \in I^c} 2m_i].$$

Proof. By the same reason as in the previous proof of Theorem 5.1.1, we know that $\Pi = i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}}$ has cohomology of finite length and is built entirely from copies of $\mathbf{1}$. Moreover, we know it is isomorphic to the direct sum of its shifted cohomologies.

Now consider the hom space $R\text{Hom}(\mathcal{L}_{\omega_d}^{\text{aut}}, \mathcal{L}_\lambda^{\text{aut}})$. By Theorem 3.3.1, we know this is isomorphic to $k[-s] \oplus k[-s-1]$, where $s = \sum_{i \in I^c} 2m_i$ is the sum appearing above. On the other hand, using Theorem 5.1.1, we can compute this as

$$R\text{Hom}(\mathcal{L}_{\omega_d}^{\text{aut}}, \mathcal{L}_\lambda^{\text{aut}}) \simeq R\text{Hom}(\mathbf{1}, \Pi) \simeq \bigoplus_i R\text{Hom}(\mathbf{1}, H^i(\Pi))[-i].$$

Comparing these calculations, it is clear that $\Pi = \pi[-s]$ for some finite-length representation π . Moreover, since π has a central character, it is necessarily a split direct sum of copies of $\mathbf{1}$. Comparing the hom calculations one more time, we see that π is of length one, so $\pi = \mathbf{1}$. This gives the result. $\square$

5.2. General points. In this section we drop the assumption that G_{b_d} is anisotropic. The explicit setting is in the following.

- Fix any $d \in \mathbb{N}$, so $G_{b_d}(E) = \text{GL}_{(d,n)}(D_{d'/n'})$.
- Let $Q \subseteq G_{b_d}$ be the evident upper triangular minimal parabolic, with block diagonal Levi M , so $M(E) = (D_{d'/n'}^\times)^{(d,n)}$.
- Let $P \subseteq G$ be the standard parabolic of which Q is an inner twist, with block diagonal Levi L . Here we implicitly choose (a representative of) b_d in $L(E)$, so $M = L_{b_d}$.

- Let $W = S_{(d,n)} = N_G(L)/L$ be the finite Weyl group of L .

Our goal is to compute the stalk $i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}}$ for any weight λ of degree d . The argument will be a combination of the arguments in the split case and the anisotropic case stated before. More precisely, we will compute the Jacquet module of $i_{b_d}^* \mathcal{L}_\lambda$ relative to Q , but since M is not a torus, the resulting analysis is considerably more complicated.

The first stages of the argument still proceed smoothly. In particular, note that the only subquotients of the stalk are the generalized Steinberg representations ρ_I . Moreover, these are separated by their Q -Jacquet modules, which can be exactly computed just as in the split case.

Proposition 5.2.1. *Let I be any d -relevant set. Then*

$$r_{G_{b_d}}^Q \rho_I = \bigoplus_{\sigma: \text{Asc}(\sigma)=I} \sigma \delta_Q^{1/2}.$$

The proof is entirely similar to the split case stated in Proposition 2.1.2(1), and we omit it. Next, to compute $r_{G_{b_d}}^Q i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}}$, we again invoke a computation of Jacquet modules by constant term functors.

Proposition 5.2.2. *We have an equivalence of functors*

$$r_{G_{b_d}}^Q i_{b_d}^* = i_{b_d}^{L*} \text{CT}_{P!}.$$

Proof. For the split case where $b_d = 1$, this is exactly [HHS24, Corollary 2.2.1 (3)]. For the general case, by the geometric second adjointness theorem (see Theorem 3.2.1 of *loc. cit.*), we obtain

$$\text{CT}_{P!} = \text{CT}_{P-*}.$$

On the other hand, write $g = (d, n)$, with $n' = n/g$ and $d' = d/g$ as before. Note that b_d has equal slope on each factor of $L \simeq \text{GL}_{n'}^g$, so along the Kottwitz map, its image on $L^{\text{ab}} \simeq \mathbb{G}_m^g$ is given by $(d', \dots, d')$; in particular, b_d is P -dominant, and hence Corollary 2.2.14 in *loc. cit.* applies. From this, we get $i_{b_d}^{L*} \text{CT}_{P-*} = r_{G_{b_d}}^Q i_{b_d}^{\text{ren}*}$. But ν_{b_d} is central as b_d is basic, and thus its centralizer $G_{\{b_d\}} = G$. In this case, we have $i_{b_d} = i_{b_d}^{\text{ren}}$, which completes the proof. $\square$

From Proposition 5.2.2 above we immediately see that

$$r_{G_{b_d}}^Q i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}} = i_{b_d}^{L*} \text{CT}_{P!} \mathcal{L}_\lambda^{\text{aut}}.$$

At this point, the analysis diverges from the split case. By Proposition 5.2.1, our next step is to take $R\text{Hom}(\sigma \delta_Q^{1/2}, -)$ for various σ and compute the resulting groups. For this, we obtain

$$\begin{aligned} R\text{Hom}(\sigma \delta_Q^{1/2}, i_{b_d}^{L*} \text{CT}_{P!} \mathcal{L}_\lambda^{\text{aut}}) &\simeq R\text{Hom}(i_{b_d!}^L(\sigma \delta_Q^{1/2}), \text{CT}_{P!} \mathcal{L}_\lambda^{\text{aut}}) \\ &\simeq R\text{Hom}(\mathcal{G}_\sigma, \text{CT}_P^{\text{spec}} \mathcal{L}_\lambda) \end{aligned}$$

for each σ . Here the first isomorphism above is trivially a rephrase with left adjoint; as for the second isomorphism above, we switch to the spectral side, writing $\mathcal{G}_\sigma$ as the coherent sheaf on Par_L matching $i_{b_d!}^L(\sigma \delta_Q^{1/2})$ under categorical local Langlands $\mathbf{L}_L$ for L .

Now the key observation is that $\mathcal{G}_\sigma$ is nothing really mysterious, and our results in the anisotropic case describe it completely. More precisely, for any $\sigma \in W$, we can form the Vogan stack $V_{\sigma\varphi}^L$ of the

parameter $\sigma\varphi$ relative to L .³ In this case, the parameter $\sigma\varphi$ turns out to be product of central twists of Steinberg parameters for the simple factors of L . Moreover, we obtain the following lemma.

Lemma 5.2.3. *For $\mathcal{G}_\sigma := \mathbf{L}_L(i_{b_d!}^L(\sigma\delta_Q^{1/2})) \in \text{Coh}(\text{Par}_L)$ defined above, we have*

$$\mathcal{G}_\sigma \simeq \iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d),$$

where $\iota_\sigma^L: V_{\sigma\varphi}^L \hookrightarrow \text{Par}_L$ is the natural closed immersion.

Proof. By construction, we have $L \simeq \text{GL}_{n'}^{(d,n)}$ and $L_{b_d}(E) \simeq (D_{d'/n'}^\times)^{(d,n)}$ with $(d', n') = 1$. Here each factor is in the anisotropic situation of Subsection 5.1. For each $\sigma \in W$, the $L_{b_d}(E)$ -representation $\sigma\delta_Q^{1/2}$ is an external tensor product of characters on the factors $D_{d'/n'}^\times$, whose L-parameters are exactly the blocks of $\sigma\varphi$. Applying Theorem 5.1.1 on each factor, and using the compatibility of CLLC with products for Levi subgroups, we obtain

$$\mathbf{L}_L(i_{b_d!}^L(\sigma\delta_Q^{1/2})) \simeq \iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\eta),$$

where $\eta = (\omega_{d'}, \dots, \omega_{d'}) \in \mathbb{X}^*(\hat{L})$. Now it remains to identify η with the restriction of ω_d to $\hat{L}$. But this follows immediately from the explicit formula for Steinberg weights: on each block of size n' , the coordinates of ω_d are of the form $\lceil \frac{rd'}{n'} \rceil - \lceil \frac{(r-1)d'}{n'} \rceil$, hence coincide with those of $\omega_{d'}$. $\square$

The various stacks introduced so far sit in the diagram

$$\begin{array}{ccccc} & & \text{Par}_P & & \\ & \swarrow \mathfrak{q} & & \searrow \mathfrak{p} & \\ V_{\sigma\varphi}^L & \xleftarrow{\iota_\sigma^L} & \text{Par}_L & & \text{Par}_G \xleftarrow{\iota} V_\varphi \end{array}$$

and in this notation we are trying to compute $R\text{Hom}(\iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d), \mathfrak{q}_* \mathfrak{p}^! \iota_* \mathcal{O}_{V_\varphi}(\lambda))$. The next step is to pass to formal completions. As in the split case, let X be the formal completion of Par_G along V_φ , and let X_P be its fiber product with Par_P , or equivalently the formal completion of Par_P along $\mathfrak{p}^{-1}(V_\varphi)$. The set of connected components of the stack X_P turns out to be indexed by W_G/W_L , but most of these components are irrelevant for our analysis. There is a distinguished collection of components $X_{P,\sigma}$ indexed by $\sigma \in W \subseteq W_G/W_L$, which can be defined as the formal completion along $\mathfrak{q}^{-1}(V_{\sigma\varphi})$. In particular, writing $X_{L,\sigma}$ for the completion of Par_L along $V_{\sigma\varphi}$, the natural map $X_{P,\sigma} \rightarrow \text{Par}_L$ factors over $X_{L,\sigma}$, and we get a diagram

$$\begin{array}{ccccc} & & X_{P,\sigma} & & \\ & \swarrow q_\sigma & & \searrow p_\sigma & \\ V_{\sigma\varphi}^L & \xleftarrow{\iota_\sigma^L} & X_{L,\sigma} & & X \xleftarrow{\iota} V_\varphi \end{array}$$

³For any $\sigma \in S_n = W_G$, the parameter $\sigma\varphi$ still factors through $\hat{L}$ and we can make sense of the Vogan stack $V_{\sigma\varphi}^L$ relative to L . However, for $\sigma \notin W$ these stacks will be irrelevant for our analysis.

of formal algebraic stacks.

By Lemma 5.2.3, the hom group we are computing coincides with

$$R\mathrm{Hom}_{X_{L,\sigma}}(\iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d), q_{\sigma*} p_{\sigma}^! \iota_{\sigma*} \mathcal{O}_{V_{\varphi}}(\lambda)).$$

Now just as in the split case, the map p_{σ} turns out to be a closed immersion onto a suitable $X_I \subseteq X$. However, the map q_{σ} is now playing a nontrivial role. More precisely, by a simple adjunction we can write the desired hom in a more symmetrical form as

$$R\mathrm{Hom}_{X_{P,\sigma}}(q_{\sigma}^* \iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d), p_{\sigma}^! \iota_{\sigma*} \mathcal{O}_{V_{\varphi}}(\lambda)).$$

In the split case, the first term essentially played no role, and we only needed to compute the global sections of $p_{\sigma}^! \iota_{\sigma*} \mathcal{O}(\lambda)$ to finish the analysis. But in the present case with q_{σ}^* involved, we have to work harder. It turns out that the arguments developed in the split case already give an explicit left-bounded resolution of $p_{\sigma}^! \iota_{\sigma*} \mathcal{O}(\lambda)$, and the morphism q_{σ} is mild enough that we can meaningfully compute the effect of $q_{\sigma*}$ on the resolution.

Next, we aim to build up local models of $X_{L,\sigma}$ and $X_{P,\sigma}$ that are similar to Proposition 3.1.1 and describe q_{σ} and p_{σ} in terms of these local models. To simplify the notation, we write

$$R_L := k[[u_1, \dots, u_n][v_j]_{j \in J_L} / (u_j v_j)_{j \in J_L}]$$

with the index subset

$$J_L := \{i \in [n-1] : n' \nmid i\} = \{i \in [n-1] : \delta_i = 0\} \subseteq [n-1].$$

Also, recall the setting that $W = N_G(L)/L$ denotes the finite Weyl group of L , so $W \simeq S_{(d,n)}$. For each $\sigma \in W$, its ascent set is $\mathrm{Asc}(\sigma) = \{j \in [(d,n)-1] : \sigma(j) < \sigma(j+1)\}$. We then write

$$R_{L,\sigma} := k[[u_1, \dots, u_n][v_j]_{j \in J_{L,\sigma}} / (u_j v_j)_{j \in J_{L,\sigma}}]$$

with the index subset

$$J_{L,\sigma} := J_L \sqcup n' \mathrm{Asc}(\sigma) = J_L \sqcup \{n'j : j \in \mathrm{Asc}(\sigma)\}.$$

Proposition 5.2.4. *For each $\sigma \in W$, with the notations above, we have the following.*

- (1) *There are isomorphisms of formal stacks*

$$\begin{aligned} X_{L,\sigma} &\simeq (\mathrm{Spf} R_L) / \hat{T}, \\ X_{P,\sigma} &\simeq (\mathrm{Spf} R_{L,\sigma}) / \hat{T}. \end{aligned}$$

In particular, the right side of the first isomorphism above is independent of σ . The torus $\hat{T}$ acts on each u_j trivially and on each v_j by simple root α_j .

- (2) *Write $R := \mathcal{O}_X$ with X given in Proposition 3.1.1. Then, with the local model in (1), the map $p_{\sigma} : X_{P,\sigma} \rightarrow X$ corresponds to the map of rings*

$$R \longrightarrow R_{P,\sigma},$$

given by $u_j \mapsto u_j$ and $v_j \mapsto v_j$ for $j \in J_{L,\sigma}$, and $v_j \mapsto 0$ for $j \notin J_{L,\sigma}$. In particular, the map p_{σ} induces an isomorphism

$$X_{P,\sigma} \xrightarrow{\sim} X_{J_{L,\sigma}} \subseteq X$$

where $X_{J_{L,\sigma}}$ is the closed substack of X defined by setting $v_j = 0$ for all $j \in J_{L,\sigma}^c$.

(3) With the local model in (1), the map $q_\sigma: X_{P,\sigma} \rightarrow X_{L,\sigma}$ corresponds to the map of rings

$$R_L \hookrightarrow R_{L,\sigma},$$

given by $v_j \mapsto v_j$ for $j \in J_L$.

Proof. Note that (2) and (3) are formal consequences of (1), so we only need to prove (1). For (1), we first describe $X_{L,\sigma}$ and then $X_{P,\sigma}$.

By construction, we have an isomorphism $\hat{L} \simeq \mathrm{GL}_{n'}^{(d,n)}$. Let $(\sigma\varphi)_a$ denote the projection of $\sigma\varphi$ to the a -th factor for each $1 \leq a \leq (d,n)$. Note that $\sigma \in W \simeq S_{(d,n)}$ only permutes the (d,n) blocks of size $n' = n/(d,n)$, while the restriction of φ to each such block is a scalar twist of the Steinberg parameter of rank n' . So for each $1 \leq a \leq (d,n)$, there exists a central character $\chi_{a,\sigma}: W_E \rightarrow \mathbb{G}_m \subseteq Z(\mathrm{GL}_{n'})$ such that $(\sigma\varphi)_a \simeq \chi_{a,\sigma} \otimes \varphi_{n'}$, where $\varphi_{n'} = \mathrm{diag}(| \cdot |^{(1-n')/2}, \dots, | \cdot |^{(n'-1)/2})$ is the semisimple Steinberg parameter for $\mathrm{GL}_{n'}$. In particular, since $\chi_{a,\sigma}$ is central, we have

$$\mathrm{ad}((\sigma\varphi)_a(\mathrm{Fr})) = \mathrm{ad}(\varphi_{n'}(\mathrm{Fr})).$$

With this condition, the argument of Proposition 3.1.1 (see [Hel23, Yan25]) applies verbatim on each factor $\mathrm{GL}_{n'}$ of $\hat{L}$, and shows that the formal completion of $\mathrm{Par}_{\mathrm{GL}_{n'}}$ along the closed Vogan substack $V_{(\sigma\varphi)_a}$ is

$$(\mathrm{Spf} k[[u_{a,1}, \dots, u_{a,n'}]][v_{a,1}, \dots, v_{a,n'-1}]/(u_{a,i}v_{a,i})_{1 \leq i \leq n'-1})/\hat{T}_a,$$

where $\hat{T}_a$, the a -th factor in the maximal torus of $\hat{L}$, acts trivially on the $u_{a,r}$ and through the simple roots on the $v_{a,r}$. Taking the product over $a = 1, \dots, g$ and reindexing $u_{a,r} = u_{(a-1)n'+r}$ and $v_{a,r} = v_{(a-1)n'+r}$, we obtain that

$$X_{L,\sigma} \simeq (\mathrm{Spf} R_L)/\hat{T}$$

as required. In particular, the right side is independent of σ .

We now turn to $X_{P,\sigma}$. Choose the standard block-permutation representative $w_\sigma \in N_{\hat{G}}(\hat{L})$ defined by $w_\sigma((a-1)n' + r) := (\sigma(a) - 1)n' + r$ for $1 \leq a \leq (d,n)$ and $1 \leq r \leq n'$. As in the proof of Proposition 3.1.2, we write $\hat{P}_\sigma := w_\sigma^{-1} \hat{P} w_\sigma$, and it is enough to work on the framed cover. Set

$$X_{P,\sigma}^\square := X^\square \times_X X_{P,\sigma}$$

where $X^\square = \mathrm{Spf} R$. Recall that a classical point of $X^\square$ is given by a pair (x, N) where x is the semisimple coordinate and $N = \sum_{i=1}^{n-1} v_i E_{i,i+1}$. By construction of Proposition 3.1.2, a framed point (x, N) lies in $X_{P,\sigma}^\square$ if and only if $N \in \mathrm{Lie}(\hat{P}_\sigma)$, or equivalently $w_\sigma N w_\sigma^{-1} \in \mathrm{Lie}(\hat{P})$. We then aim to compute this condition explicitly.

Claim. We have $N = \sum_{i=1}^{n-1} v_i E_{i,i+1} \in \mathrm{Lie}(\hat{P}_\sigma)$ if and only if $v_i = 0$ for all $i \in J_{L,\sigma}^c$.

Proof of Claim. If $i \in J_L$, then $i = (a-1)n' + r$ for some $1 \leq r \leq n' - 1$, hence

$$w_\sigma E_{i,i+1} w_\sigma^{-1} = E_{(\sigma(a)-1)n'+r, (\sigma(a)-1)n'+r+1},$$

which lies inside a single diagonal block of $\hat{P}$. Therefore, given any $\sigma \in S_{(d,n)}$, we have $E_{i,i+1} \in \mathrm{Lie}(\hat{P}_\sigma)$ for all $i \in J_L$. On the other hand, if $i \notin J_L$, or equivalently

$i = an'$ with $1 \leq a \leq (d, n) - 1$, then

$$w_\sigma E_{an', an'+1} w_\sigma^{-1} = E_{\sigma(a)n', (\sigma(a+1)-1)n'+1}.$$

The right side above belongs to $\text{Lie}(\hat{P})$ if and only if the $\sigma(a)$ -th block lies above the $\sigma(a+1)$ -th block in the standard block upper-triangular order, equivalently if and only if $\sigma(a) < \sigma(a+1)$. Thus we see $E_{an', an'+1} \in \text{Lie}(\hat{P}_\sigma)$ if and only if $a \in \text{Asc}(\sigma)$. On the other hand, we have $J_{L, \sigma} = J_L \sqcup n' \text{Asc}(\sigma)$ by definition; combining the two cases above, the desired claim follows. $\square$

From the claim, we conclude that the image of the framed projection $p_\sigma^\square$ is exactly $\text{Spf } R_{L, \sigma}$ where $R_{L, \sigma} = R/(v_i)_{i \in J_{L, \sigma}^c}$ by definition; in other words, this framed projection factors as

$$p_\sigma^\square: X_{P, \sigma}^\square \longrightarrow \text{Spf } R_{L, \sigma} \hookrightarrow X^\square.$$

On the other hand, if $(x, N) \in \text{Spf } R_{L, \sigma}$, then the previous calculation shows that $N \in \text{Lie}(\hat{P}_\sigma)$, hence the datum $(x, N; \hat{P}_\sigma)$ defines a point of $X_{P, \sigma}^\square$. As a consequence,

$$X_{P, \sigma}^\square \xrightarrow{\sim} \text{Spf } R_{L, \sigma}.$$

Passing to the quotient by $\hat{T}$, we obtain the desired isomorphism

$$X_{P, \sigma} \simeq (\text{Spf } R_{L, \sigma})/\hat{T}.$$

This finishes the proof of part (1), and then (2) and (3) directly follow from (1). $\square$

Theorem 5.2.5. *Suppose $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ is any weight of degree d . Then for any $\sigma \in W \simeq S_{(d, n)}$, the complex*

$$R\text{Hom}_{X_{P, \sigma}}(q_\sigma^* \iota_{\sigma*}^L \mathcal{O}_{V_\varphi^L}(\omega_d), p_\sigma^! \iota_{\sigma*} \mathcal{O}_{V_\varphi}(\lambda))$$

carries a natural right action of $R\text{Hom}_{X_{L, \sigma}}(\iota_{\sigma}^L \mathcal{O}_{V_\varphi^L}(\omega_d), \iota_{\sigma*}^L \mathcal{O}_{V_\varphi^L}(\omega_d))$ from pre-composition in the first variable. Moreover, this right module vanishes unless $I_\lambda = \text{Asc}(\sigma)$, and in the non-vanishing case it is isomorphic to*

$$\begin{aligned} R\text{Hom}_{X_{L, \sigma}}(\iota_{\sigma*}^L \mathcal{O}_{V_\varphi^L}(\omega_d), \iota_{\sigma*}^L \mathcal{O}_{V_\varphi^L}(\omega_d))[s(\lambda)] &\simeq \wedge^\bullet k^{(d, n)}[s(\lambda)] \\ &\simeq \text{Ext}_{M(E)}^\bullet(\sigma \delta_Q^{1/2}, \sigma \delta_Q^{1/2})[s(\lambda)]. \end{aligned}$$

Here, the subsets $I_\lambda = \{i \in [(d, n) - 1]: m_{n'i} \leq 0\}$ and $\text{Asc}(\sigma) = \{i \in [(d, n) - 1]: \sigma(i) < \sigma(i+1)\}$ are as usual; the degree is

$$s(\lambda) := \sum_{i \in [n-1]: m_i > 0} (\delta_i - 2m_i)$$

with $\delta_i \in \{0, 1\}$, and $\delta_i = 1$ if and only if $n' \mid i$.

For our purpose of deducing Theorem 1.2.1 from Theorem 5.2.5 above, we need to recognize the complex $\text{Ext}_{M(E)}^\bullet(\sigma \delta_Q^{1/2}, \sigma \delta_Q^{1/2})$ via $R\text{Hom}(q_\sigma^* \iota_{\sigma*}^L \mathcal{O}_{V_\varphi^L}(\omega_d), p_\sigma^! \iota_{\sigma*} \mathcal{O}_{V_\varphi}(\lambda))$. Notice that in the case where $\text{Asc}(\sigma) = I_\lambda$, the cohomology of the left-hand $R\text{Hom}$ in degree $r + s(\lambda)$ is a k -vector space of dimension $\binom{(d, n)}{r}$. The point is that $\text{Ext}_{M(E)}^\bullet(\sigma \delta_Q^{1/2}, \sigma \delta_Q^{1/2})$ is an exterior algebra on a k -vector space of dimension (d, n) .

5.3. Proof of Theorem 1.2.1. We now deduce our main result Theorem 1.2.1 from Theorem 5.2.5. Assume throughout the Eisenstein compatibility. Fix a weight $\lambda = \omega_d + \sum_{1 \leq i \leq n-1} m_i \alpha_i$ of degree d , and set

$$\Pi := i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}} \in \mathcal{D}(G_{b_d}(E), k).$$

By Proposition 2.3.1, the cohomology objects of Π have finite length, and every irreducible subquotient is some generalized Steinberg representation $\rho_{\mathbf{I}}$.

For the minimal parabolic $Q \subseteq G_{b_d}$ with Levi M from Subsection 5.2, consider the projection of the Jacquet module

$$\mathcal{D}(M(E), k) \longrightarrow \mathfrak{B}_\sigma, \quad r_{G_{b_d}}^Q \Pi \longmapsto (r_{G_{b_d}}^Q \Pi)_\sigma$$

where $\mathfrak{B}_\sigma$ is the Bernstein block of $M(E)$ containing $\sigma\delta_Q^{1/2}$. Since for varying σ , the characters $\sigma\delta_Q^{1/2}$ are pairwise non-isomorphic, we have

$$r_{G_{b_d}}^Q \Pi \simeq \bigoplus_{\sigma \in S_n} (r_{G_{b_d}}^Q \Pi)_\sigma,$$

and

$$R\text{Hom}_{M(E)}(\sigma\delta_Q^{1/2}, r_{G_{b_d}}^Q \Pi) \simeq R\text{Hom}_{M(E)}(\sigma\delta_Q^{1/2}, (r_{G_{b_d}}^Q \Pi)_\sigma).$$

By Proposition 5.2.2, Lemma 5.2.3, and passage to the formal completion at the point $\sigma\varphi$, the right side above is further isomorphic to

$$R\text{Hom}_{X_{P,\sigma}}(q_\sigma^* \iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d), p_\sigma^! \iota_{\sigma*} \mathcal{O}_{V_\varphi}(\lambda)).$$

Recall from Lemma 5.2.3 that under categorical local Langlands correspondence for L , the coherent sheaf $\iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d)$ on $X_{L,\sigma}$ corresponds to $i_{b_d!}^L(\sigma\delta_Q^{1/2})$. By the full faithfulness of $\mathbf{L}_L$, it then follows that

$$R\text{Hom}_{X_{L,\sigma}}(\iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d), \iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d)) \simeq \text{Ext}_{M(E)}^\bullet(\sigma\delta_Q^{1/2}, \sigma\delta_Q^{1/2}).$$

On the other hand, by adjunction we have

$$R\text{Hom}_{X_{P,\sigma}}(q_\sigma^* \iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d), p_\sigma^! \iota_{\sigma*} \mathcal{O}_{V_\varphi}(\lambda)) \simeq R\text{Hom}_{X_{L,\sigma}}(\iota_{\sigma*}^L \mathcal{O}_{V_{\sigma\varphi}^L}(\omega_d), q_{\sigma*} p_\sigma^! \iota_{\sigma*} \mathcal{O}_{V_\varphi}(\lambda)),$$

so the left side above carries a natural structure of right module over $\text{Ext}_{M(E)}^\bullet(\sigma\delta_Q^{1/2}, \sigma\delta_Q^{1/2})$. Combining this with Theorem 5.2.5, we see that

$$R\text{Hom}_{M(E)}(\sigma\delta_Q^{1/2}, (r_{G_{b_d}}^Q \Pi)_\sigma) \simeq \begin{cases} \text{Ext}_{M(E)}^\bullet(\sigma\delta_Q^{1/2}, \sigma\delta_Q^{1/2})[s(\lambda)], & \mathbf{I}_\lambda = \text{Asc}(\sigma), \\ 0, & \mathbf{I}_\lambda \neq \text{Asc}(\sigma), \end{cases}$$

as right $\text{Ext}_{M(E)}^\bullet(\sigma\delta_Q^{1/2}, \sigma\delta_Q^{1/2})$ -modules. Now we claim that the result above further implies

$$(r_{G_{b_d}}^Q \Pi)_\sigma \simeq \begin{cases} \sigma\delta_Q^{1/2}[s(\lambda)], & \mathbf{I}_\lambda = \text{Asc}(\sigma), \\ 0, & \mathbf{I}_\lambda \neq \text{Asc}(\sigma). \end{cases}$$

Indeed, to verify this, we use the structure of the Bernstein block $\mathfrak{B}_\sigma$. Exactly as in the proof of Theorem 5.1.1, but now factorwise, the abelian category of finite-length smooth $M(E)$ -representations

all of whose irreducible subquotients are isomorphic to $\sigma\delta_Q^{1/2}$ is equivalent to the category of finite-length modules over the regular local ring

$$R_\sigma := k[T_1^{\pm 1}, \dots, T_{(d,n)}^{\pm 1}]_{\mathfrak{m}_\sigma},$$

where $\mathfrak{m}_\sigma$ is the maximal ideal corresponding to $\sigma\delta_Q^{1/2}$. Under this equivalence, $\sigma\delta_Q^{1/2}$ corresponds to the residue field $R_\sigma/\mathfrak{m}_\sigma$. Since R_σ is a regular local ring of dimension (d, n) , standard Koszul duality yields an equivalence

$$R\mathrm{Hom}_{M(E)}(\sigma\delta_Q^{1/2}, -) : D_{\mathrm{fl}}^b(\mathfrak{B}_\sigma) \xrightarrow{\sim} \mathrm{Perf}(\mathrm{Ext}_{M(E)}^\bullet(\sigma\delta_Q^{1/2}, \sigma\delta_Q^{1/2})).$$

Applying this equivalence, the claim about $(r_{G_{b_d}}^Q \Pi)_\sigma$ directly follows. Summing over $\sigma \in S_n$, we then obtain

$$r_{G_{b_d}}^Q \Pi \simeq \bigoplus_{\sigma: \mathrm{Asc}(\sigma)=I_\lambda} \sigma\delta_Q^{1/2}[s(\lambda)].$$

On the other hand, Proposition 5.2.1 gives

$$r_{G_{b_d}}^Q \rho_{I_\lambda} = \bigoplus_{\sigma: \mathrm{Asc}(\sigma)=I_\lambda} \sigma\delta_Q^{1/2}.$$

Comparing the two descriptions above, we conclude that

$$r_{G_{b_d}}^Q \Pi \simeq r_{G_{b_d}}^Q \rho_{I_\lambda}[s(\lambda)].$$

Taking cohomology objects, since the Jacquet module functor $r_{G_{b_d}}^Q$ is exact, this implies

$$H^i(r_{G_{b_d}}^Q \Pi) = r_{G_{b_d}}^Q (H^i(\Pi)).$$

But the object above vanishes for all degrees $i \neq -s(\lambda)$, because $r_{G_{b_d}}^Q \Pi \simeq r_{G_{b_d}}^Q \rho_{I_\lambda}[s(\lambda)]$ is concentrated in degree $-s(\lambda)$. Also, we claim that the Jacquet module functor is conservative on the finite-length subcategory generated by the generalized Steinberg representations ρ_I . Indeed, if ϖ is any nonzero object in this category, then ϖ has some irreducible subquotient ρ_I , so $r_{G_{b_d}}^Q \varpi$ has subquotient $r_{G_{b_d}}^Q \rho_I$, which is nonzero by exactness.

Next, since the Jacquet module functor is exact and conservative, we have for all $i \neq -s(\lambda)$ that $H^i(\Pi) = 0$. Now for our purpose, it suffices to consider $H^{-s(\lambda)}(\Pi)$. From the computation above, it is clear that

$$r_{G_{b_d}}^Q (H^{-s(\lambda)}(\Pi)) \simeq \bigoplus_{\sigma: \mathrm{Asc}(\sigma)=I_\lambda} \sigma\delta_Q^{1/2} = r_{G_{b_d}}^Q \rho_{I_\lambda}.$$

Now Proposition 5.2.1 shows that for varying subsets I , the character sets appearing in the right side above are pairwise disjoint. But by construction, every irreducible subquotient of $H^{-s(\lambda)}(\Pi)$ is some ρ_I , so the above equality forces $H^{-s(\lambda)}(\Pi)$ to have exactly one Jordan–Hölder constituent, namely ρ_{I_λ} , with multiplicity one. Therefore,

$$H^{-s(\lambda)}(\Pi) \simeq \rho_{I_\lambda}.$$

So we conclude that

$$i_{b_d}^* \mathcal{L}_\lambda^{\text{aut}} = \Pi \simeq \rho_{I_\lambda}[s(\lambda)],$$

which is exactly Conjecture 1.1.2. This finishes the proof of our main result Theorem 1.2.1.

6. CANONICAL EXTENSIONS

6.1. Construction and first properties. For a fixed I , consider the ind-coherent sheaf

$$\begin{aligned} \mathcal{F}_I &= \operatorname{colim}_{\min m_i \rightarrow \infty} \mathcal{L}_{-\sum_{i \in I} m_i \alpha_i + \sum_{i \in I^c} m_i \alpha_i}[-|I^c| + 2 \sum_{i \in I^c} m_i] \\ &\in \operatorname{IndCoh}(\operatorname{Par}_{\operatorname{GL}_n}) \end{aligned}$$

where the colimit is taken along the evident compositions of lowering operators $L_{\alpha_i} : \mathcal{L}_\lambda \rightarrow \mathcal{L}_{\lambda - \alpha_i}$ and raising operators $R_{\alpha_i} : \mathcal{L}_\lambda \rightarrow \mathcal{L}_{\lambda + \alpha_i}[2]$.

Assuming strong categorical local Langlands, we can define the associated automorphic sheaf $\mathcal{F}_I^{\text{aut}}$. In what follows, any time we make a statement about the latter sheaf, we are tacitly assuming strong CLLC (and hence the Eisenstein compatibility, as in Conjecture 2.2.2).

Before stating properties about $\mathcal{F}_I$, we introduce some simplified notations as follows.

Notation 6.1.1. Fix a subset $I \subseteq [n-1]$. Let $\mathbf{k} = (k_1, \dots, k_{n-1}) \in \mathbb{Z}^{n-1}$ be an $(n-1)$ -tuple of integers. Recall from 1.5.4 that the weight $\lambda_{\mathbf{k}, I}$ and its transpose are defined respectively as

- $\lambda_{I, \mathbf{k}} := -\sum_{i \in I} k_i \alpha_i + \sum_{i \in I^c} k_i \alpha_i,$
- $\lambda_{I, \mathbf{k}}^\dagger := -\sum_{i \in I} k_i \alpha_{n-i} + \sum_{i \in I^c} k_i \alpha_{n-i}.$

Note that $\lambda_{I, \mathbf{k}}$ and $\lambda_{I, \mathbf{k}}^\dagger$ are of degree zero, and we always have $\lambda_{I, \mathbf{k}}^\dagger = \lambda_{I^t, \mathbf{k}^t}$ with $\mathbf{k}^t = (k_{n-1}, \dots, k_1)$. Then for $(n-1)$ -tuples $\mathbf{r}$ and $\mathbf{s}$, we set

$$\begin{aligned} \mathcal{F}_{I, \mathbf{r}} &:= \mathcal{L}_{\lambda_{I, \mathbf{r}}}[-|I^c| + 2 \sum_{i \in I^c} r_i], \\ \mathcal{F}_{I, \mathbf{s}}^\dagger &:= \mathcal{L}_{\lambda_{I, \mathbf{s}}^\dagger}[-|I^c| + 2 \sum_{i \in I^c} s_i]. \end{aligned}$$

With these notations, we get inductive systems $\{\mathcal{F}_{I, \mathbf{r}}\}_{\mathbf{r}}$ and $\{\mathcal{F}_{I, \mathbf{s}}^\dagger\}_{\mathbf{s}}$, in which $\mathbf{r} \rightarrow \infty$ and $\mathbf{s} \rightarrow \infty$ mean respectively that $\min\{r_i\} \rightarrow \infty$ and $\min\{s_i\} \rightarrow \infty$. Then we have

$$\mathcal{F}_I = \operatorname{colim}_{\mathbf{r}} \mathcal{F}_{I, \mathbf{r}}, \quad \mathcal{F}_{I^t} = \operatorname{colim}_{\mathbf{s}} \mathcal{F}_{I, \mathbf{s}}^\dagger.$$

Here, the first equality follows from definition of $\mathcal{F}_I$; to check the second equality, one only needs to observe that $\mathcal{F}_{I, \mathbf{s}}^\dagger = \mathcal{F}_{I^t, \mathbf{s}^t}$ together with that $\mathbf{s} \rightarrow \infty$ is equivalent to $\mathbf{s}^t \rightarrow \infty$.

Proposition 6.1.2. *Fix a subset $I \subseteq [n-1]$ as above.*

- (1) *We have $i_1^* \mathcal{F}_I^{\text{aut}} = \pi_I$.*
- (2) *The sheaf $\mathcal{F}_I$ is an admissible ind-coherent sheaf. In particular, $\mathcal{F}_I^{\text{aut}}$ is ULA with stalks of finite length.*

Proof. (1) By construction of $\mathcal{F}_I$, we have $i_1^* \mathcal{F}_I = i_1^*(\operatorname{colim}_{\mathbf{r}} \mathcal{F}_{I, \mathbf{r}}) = \operatorname{colim}_{\mathbf{r}} i_1^* \mathcal{F}_{I, \mathbf{r}}$. Since the strong CLLC is assumed, applying Theorem 1.2.1 to $i_1^* \mathcal{F}_{I, \mathbf{r}}$ immediately gives the assertion.

(2) Granting the first claim, the second claim is a formal consequence of CLLC. For the first claim about the admissibility, it suffices to check that for $\mathcal{G}$ varying over a set of compact generators of $\mathrm{IndCoh}(\mathrm{Par}_G^{\mathrm{unip}})$, the derived complex $R\mathrm{Hom}(\mathcal{G}, \mathcal{F}_I)$ is a perfect complex of k -vector spaces.

More precisely, we will verify this for $\mathcal{G}$ varying in the set of generators given by the twisted coherent Springer sheaves $\mathcal{S}_\lambda = \mathfrak{p}_* \mathfrak{q}^* \mathcal{O}_{\mathrm{Par}_T^{\mathrm{unr}}}(\lambda)$ with $\lambda \in \mathbb{X}^*(\hat{T})$. The functor $R\mathrm{Hom}(\mathcal{S}_\lambda, -)$ preserves filtered colimits by compactness, so it suffices to consider

$$R\mathrm{Hom}(\mathcal{S}_\lambda, \mathcal{F}_I) \simeq \mathrm{colim}_r R\mathrm{Hom}(\mathcal{S}_\lambda, \mathcal{F}_{I,r}).$$

We can further reduce this $R\mathrm{Hom}$ from Par_G to X . Let $\varepsilon: X \rightarrow \mathrm{Par}_G$ be the evident natural map. Recall that we slightly abusively view $\mathcal{L}_{\lambda_{I,r}}$ as a sheaf on X , so we identify $\mathcal{F}_{I,r}$ on X with $\varepsilon_* \mathcal{F}_{I,r}$ on Par_G . Then the $R\mathrm{Hom}$ in the colimit above is

$$R\mathrm{Hom}_{\mathrm{Par}_G}(\mathcal{S}_\lambda, \varepsilon_* \mathcal{F}_{I,r}) \simeq R\mathrm{Hom}_X(\varepsilon^* \mathcal{S}_\lambda, \mathcal{F}_{I,r}).$$

To further compute $\varepsilon^* \mathcal{S}_\lambda$, by construction in Subsection 3.1 and Proposition 3.1.2, writing $\tilde{\varepsilon}: \tilde{X} \rightarrow \mathrm{Par}_B$ the base change of $\varepsilon: X \rightarrow \mathrm{Par}_G$ along $\mathfrak{p}: \mathrm{Par}_B \rightarrow \mathrm{Par}_G$, the connected components of $\tilde{X}$ are exactly labelled by $\sigma \in S_n$, and we obtain the following.

$$\begin{array}{ccccc}
 \tilde{X}_\sigma & & & & \\
 \downarrow \mathrm{Id} & \searrow p_\sigma & & & \\
 \tilde{X} & & X_{\mathrm{Asc}(\sigma)} & & \\
 \downarrow \tilde{\varepsilon} & \searrow p & \downarrow \mathrm{Id} & & \\
 & & X & & \\
 & \square & \downarrow \varepsilon & & \\
 \mathrm{Par}_B & \xrightarrow{\mathfrak{p}} & \mathrm{Par}_G & & \\
 \nwarrow \mathfrak{q} & & & & \\
 \mathrm{Par}_T & & & &
 \end{array}$$

In the Cartesian diagram, note that ε is of finite tor-amplitude, as Par_G is an Artin stack locally of finite type and V_φ is locally defined by the ideal that is finitely generated by u_i 's. With this finiteness, we apply [Zhu25, Lemma 9.26 (3)], which states that the Beck–Chevalley map is an isomorphism, i.e., $\varepsilon^* \mathfrak{p}_* \simeq \mathfrak{p}_* \tilde{\varepsilon}^*$ as functors on IndCoh . Thus, it follows that

$$\varepsilon^* \mathcal{S}_\lambda \simeq \mathfrak{p}_* \tilde{\varepsilon}^* \mathfrak{q}^* \mathcal{O}_{\mathrm{Par}_T^{\mathrm{unr}}}(\lambda) \simeq \bigoplus_{\sigma \in S_n} \mathcal{O}_{X_{\mathrm{Asc}(\sigma)}}(\sigma \lambda),$$

where by our usual convention, we view each $\mathcal{O}_{X_{\mathrm{Asc}(\sigma)}}$ as a sheaf on X by omitting the evident pushforward. To verify the $\hat{T}$ -weight on the right side of this decomposition, recall that the left action of S_n on $\lambda \in \mathbb{X}^*(\hat{T})$ is defined by $(\sigma \lambda)(t) = \lambda(w_\sigma^{-1} t w_\sigma)$; but in the proof of Proposition 3.1.2, on each component labelled by σ the universal Borel is $\hat{B}_\sigma = w_\sigma^{-1} \hat{B} w_\sigma$.

Therefore, we compute

$$\begin{aligned}
R\mathrm{Hom}(\mathcal{S}_\lambda, \mathcal{F}_I) &\simeq \operatorname{colim}_{\mathbf{r}} R\mathrm{Hom}_X(\varepsilon^* \mathcal{S}_\lambda, \mathcal{F}_{I, \mathbf{r}}) \\
&\simeq \operatorname{colim}_{\mathbf{r}} \bigoplus_{\sigma \in S_n} R\mathrm{Hom}_X(\mathcal{O}_{X_{\mathrm{Asc}(\sigma)}}(\sigma \lambda), \mathcal{F}_{I, \mathbf{r}}) \\
&\simeq \operatorname{colim}_{\mathbf{r}} \bigoplus_{\sigma \in S_n} R\mathrm{Hom}_X(\mathcal{O}_{X_{\mathrm{Asc}(\sigma)}}, \mathcal{L}_{\lambda_{I, \mathbf{r}} - \sigma \lambda})[-|\mathrm{I}^c| + 2 \sum_{i \in \mathrm{I}^c} r_i].
\end{aligned}$$

Notice that $\lambda_{I, \mathbf{r}} - \sigma \lambda$ is of degree zero, so we rewrite it as $\sum_{1 \leq i \leq n-1} t_i \alpha_i$. By Theorem 3.2.1, each $R\mathrm{Hom}_X$ in the last line above vanishes unless $\mathrm{Asc}(\sigma) = \{i \in [n-1] : t_i \leq 0\}$. As $\sigma \lambda$ is independent of $\mathbf{r}$, the set $\{i \in [n-1] : t_i \leq 0\}$ keeps invariant whenever $\min\{r_i\}$ becomes sufficiently large. Thus, using Theorem 3.2.1 again, the direct sum in the colimit above stabilizes as $\mathbf{r} \rightarrow \infty$ and converges to finitely many copies of 1-dimensional k -vector spaces with shifts. In this situation, each transition map on $R\mathrm{Hom}_X$'s, induced by raising and lowering operators, is either zero or an isomorphism; here, note that each $R\mathrm{Hom}$ is concentrated in two consecutive degrees, but along $R_{\alpha_i} : \mathcal{L}_\lambda \rightarrow \mathcal{L}_{\lambda + \alpha_i}[2]$, shifting the degree by 2 forces the transition map to be zero.

To conclude, we have shown that $R\mathrm{Hom}(\mathcal{S}_\lambda, \mathcal{F}_I)$ is perfect for each $\lambda \in \mathbb{X}^*(\hat{T})$, and thus the admissibility of $\mathcal{F}_I$ follows. $\square$

The argument above actually gives strong information about the stalks of $\mathcal{F}_I^{\mathrm{aut}}$ at all split points. More precisely, let μ be a dominant character of degree zero, viewed as a character into G , and let $b_\mu = \mu(\varpi)$ be the evident element of $B(G)$. Let P_μ be the attracting parabolic of μ with Levi M_μ . Then M_μ is canonically identified with G_{b_μ} , and in particular the latter group is split.

Proposition 6.1.3. *For any dominant μ of degree zero, we have*

$$[i_{b_\mu}^{*\mathrm{ren}} \mathcal{F}_I^{\mathrm{aut}}] = [r_G^{P_\mu^-} \pi_I], \quad [i_{b_\mu}^{\mathrm{ren}} \mathcal{F}_I^{\mathrm{aut}}] = [r_G^{P_\mu} \pi_I],$$

where $[-]$ indicates the image in the Grothendieck group of finite-length $M_\mu(E)$ -representations.

Proof. From Proposition 6.1.2, we see $i_1^* \mathcal{F}_I^{\mathrm{aut}} = \pi_I$, and both stalks $i_{b_\mu}^{*\mathrm{ren}} \mathcal{F}_I^{\mathrm{aut}}$ and $i_{b_\mu}^{\mathrm{ren}} \mathcal{F}_I^{\mathrm{aut}}$ are of finite length. To get the first equality, it suffices to check the equivalence of functors, for which we directly compute

$$r_G^{P_\mu^-} i_1^* = i_1^{M_\mu^*} \mathrm{CT}_{P_\mu!} = i_{b_\mu}^{*\mathrm{ren}}.$$

To apply the results about the geometric constant term in [HHS24], notice that for such μ , we have $M_\mu = G_{\{b_\mu\}}$ and $P_\mu = P_{\{b_\mu\}}^-$ by construction. The first identification above is due to Corollary 2.2.1 (3) of *loc. cit.*, where

$$i_1^{M_\mu} : [* / M_\mu(E)] \xrightarrow{\sim} \mathrm{Bun}_{M_\mu}^{b_\mu} \hookrightarrow \mathrm{Bun}_{M_\mu}.$$

Here the subscript 1 in $i_1^{M_\mu}$ denotes the unique point of $[* / M_\mu(E)]$, and one may rename this point b_μ to match the usual convention of [Han23]. With this notation, the second identification $i_1^{M_\mu^*} \mathrm{CT}_{P_\mu!} = i_{b_\mu}^{*\mathrm{ren}}$ is given by [HHS24, Corollary 2.2.5 (4)]. This deduces the first equality, and the second can be shown similarly. $\square$

6.2. Maps between canonical extensions. Motivated by Subsection 3.3, we consider maps between $\mathcal{F}_I$'s and prove the following result, which can be regarded as a colimit version of Theorem 3.3.1.

Proposition 6.2.1. *For arbitrary subsets $I, J \subseteq [n-1]$, their associated canonical extensions $\mathcal{F}_I, \mathcal{F}_J \in \text{IndCoh}(\text{Par}_G)$ satisfy that*

$$R\text{Hom}(\mathcal{F}_I, \mathcal{F}_J) = \begin{cases} k \oplus k[-1], & I = J, \\ 0, & I \neq J. \end{cases}$$

Proof. As in Notation 6.1.1, we write $\mathcal{F}_I = \text{colim}_{\mathbf{r}} \mathcal{F}_{I,\mathbf{r}}$ and $\mathcal{F}_J = \text{colim}_{\mathbf{s}} \mathcal{F}_{J,\mathbf{s}}$, where each $\mathcal{F}_{I,\mathbf{r}}$ or $\mathcal{F}_{J,\mathbf{s}}$ is a coherent sheaf, and hence a compact object of $\text{IndCoh}(\text{Par}_G)$. Then we obtain

$$R\text{Hom}(\mathcal{F}_I, \mathcal{F}_J) = \lim_{\mathbf{r}} \text{colim}_{\mathbf{s}} R\text{Hom}(\mathcal{F}_{I,\mathbf{r}}, \mathcal{F}_{J,\mathbf{s}}).$$

Applying Theorem 3.3.1, given $\mathbf{r}, \mathbf{s}$, there exists some integer $A_{\mathbf{r},\mathbf{s}}$ such that

$$R\text{Hom}(\mathcal{F}_{I,\mathbf{r}}, \mathcal{F}_{J,\mathbf{s}}) \simeq k[A_{\mathbf{r},\mathbf{s}}] \oplus k[-1 + A_{\mathbf{r},\mathbf{s}}].$$

Here $A_{\mathbf{r},\mathbf{s}}$ can be computed explicitly, which also depends on I and J . From the proof of Proposition 6.1.2 (2), we see that as $\mathbf{s} \rightarrow \infty$, each transition map between $R\text{Hom}$'s becomes either zero or an isomorphism, and hence $R\text{Hom}(\mathcal{F}_{I,\mathbf{r}}, \mathcal{F}_{J,\mathbf{s}})$ stabilizes with respect to $\mathbf{s}$. More precisely, it is easy to compute

$$\text{colim}_{\mathbf{s}} R\text{Hom}(\mathcal{F}_{I,\mathbf{r}}, \mathcal{F}_{J,\mathbf{s}}) \simeq k[A_{\mathbf{r}}] \oplus k[-1 + A_{\mathbf{r}}]$$

with $A_{\mathbf{r}} = |\mathbb{I}^c| - |\mathbb{J}^c| - 2 \sum_{i \in (\mathbb{I} \cap \mathbb{J}^c) \cup (\mathbb{I}^c \cup \mathbb{J})} r_i$.

To attain the limit of this pro-system with respect to $\mathbf{r}$, consider replacing $\mathbf{r}$ with $\mathbf{r} + \mathbf{e}_i$, and the transition map (up to isomorphism) is written as

$$k[A_{\mathbf{r}+\mathbf{e}_i}] \oplus k[-1 + A_{\mathbf{r}+\mathbf{e}_i}] \longrightarrow k[A_{\mathbf{r}}] \oplus k[-1 + A_{\mathbf{r}}].$$

But we notice that

$$A_{\mathbf{r}+\mathbf{e}_i} - A_{\mathbf{r}} = \begin{cases} -2, & \text{if } i \in (\mathbb{I} \cap \mathbb{J}^c) \cup (\mathbb{I}^c \cup \mathbb{J}), \\ 0, & \text{otherwise.} \end{cases}$$

Similarly, as each complex is concentrated in only two consecutive degrees, shifting by even integers gives the zero transition map, and finally deduces the zero limit. Finally, note that $(\mathbb{I} \cap \mathbb{J}^c) \cup (\mathbb{I}^c \cup \mathbb{J}) = \emptyset$ if and only if $I = J$, so the desired result follows. $\square$

6.3. Duality. In this section we prove the following theorem.

Theorem 6.3.1. *Fix any subset $I \subseteq [n-1]$. Then*

$$R\text{Hom}(\mathcal{F}_{I^t}, \mathbb{D}_{\text{tw.adm}}(\mathcal{F}_I)) \simeq k \oplus k[-1].$$

Choosing any nontrivial class in degree zero, the resulting map $\gamma: \mathcal{F}_{I^t} \xrightarrow{\sim} \mathbb{D}_{\text{tw.adm}}(\mathcal{F}_I)$ is an isomorphism.

Proof. Some notations in the following have been defined in Notation 6.1.1. To tackle with the desired $R\text{Hom}$, we compute

$$\begin{aligned} R\text{Hom}(\mathcal{F}_{\text{I}^t}, \mathbb{D}_{\text{tw.adm}}(\mathcal{F}_{\text{I}})) &= \lim_s \text{colim}_r R\text{Hom}(\mathcal{F}_{\text{I},s}^\dagger, \mathbb{D}_{\text{tw.adm}}(\mathcal{F}_{\text{I},r})) \\ &= \lim_s \text{colim}_r R\text{Hom}(\mathbb{D}_{\text{tw.GS}}(\mathcal{F}_{\text{I},s}^\dagger), \mathcal{F}_{\text{I},r})^\vee \\ &=: \lim_s \text{colim}_r \mathcal{A}_{s,r}^\vee \end{aligned}$$

where in the second line we used the duality exchange formula. Now for any fixed $\mathbf{s}$, along the dual weight map $\lambda \mapsto \lambda^*$ defined before Proposition 2.3.4, we have

$$(\lambda_{\text{I},s}^\dagger)^* = \lambda_{[n-1],\mathbf{1}} - \lambda_{\text{I}^t,s}^\dagger = \lambda_{[n-1],\mathbf{1}} - \lambda_{\text{I},s}.$$

Using Proposition 2.3.4 on the spectral side, i.e., $\mathbb{D}_{\text{tw.GS}}(\mathcal{L}_\lambda) = \mathcal{L}_{\lambda^*}[-1]$ for any weight λ , it is easy to see that

$$\mathbb{D}_{\text{tw.GS}}(\mathcal{F}_{\text{I},s}^\dagger) \simeq \mathcal{L}_{\lambda_{[n-1],\mathbf{1}} - \lambda_{\text{I},s}}[|\text{I}^c| - 2 \sum_{i \in \text{I}^c} s_i - 1],$$

where $\mathbf{1}$ is the $(n-1)$ -tuple of 1. So the previous $R\text{Hom}$ in the limit is further calculated as

$$\begin{aligned} \mathcal{A}_{s,r} &\simeq R\text{Hom}(\mathcal{L}_{\lambda_{[n-1],\mathbf{1}} - \lambda_{\text{I},s}}[|\text{I}^c| - 2 \sum_{i \in \text{I}^c} s_i - 1], \mathcal{L}_{\lambda_{\text{I},r}}[-|\text{I}^c| + 2 \sum_{i \in \text{I}^c} r_i]) \\ &= R\text{Hom}(\mathcal{L}_{\lambda_{[n-1],\mathbf{1}} - \lambda_{\text{I},s}}, \mathcal{L}_{\lambda_{\text{I},r}})[1 + 2 \sum_{i \in \text{I}^c} (s_i + r_i - 1)]. \end{aligned}$$

Now we compute that

$$(\lambda_{[n-1],\mathbf{1}} - \lambda_{\text{I},s}) - \lambda_{\text{I},r} = \sum_{i \in \text{I}} (1 + s_i + r_i) \alpha_i + \sum_{i \in \text{I}^c} (1 - s_i - r_i) \alpha_i.$$

Then by carefully applying Theorem 3.3.1, the shifted $R\text{Hom}$ above simplifies to $k \oplus k[1]$. Finally, dualizing again, we see that

$$\mathcal{A}_{s,r}^\vee = R\text{Hom}(\mathbb{D}_{\text{tw.GS}}(\mathcal{F}_{\text{I},s}^\dagger), \mathcal{F}_{\text{I},r})^\vee \simeq k \oplus k[-1].$$

This shows that the terms in the system $\{\mathcal{A}_{s,r}\}$ are constant with respect to $\mathbf{s}$ and $\mathbf{r}$. Moreover, as the transition maps are given in terms of suitable lowering and raising maps, it is easy to see that they are isomorphisms, so passing to the limit gives the claimed result that $\gamma: \mathcal{F}_{\text{I}^t} \xrightarrow{\sim} \mathbb{D}_{\text{tw.adm}}(\mathcal{F}_{\text{I}})$ is an isomorphism. $\square$

Recall that we have conjectured in Subsection 1.4 that $\mathbb{D}_{\text{Verd}} \mathcal{F}_{\text{I}}^{\text{aut}} = \mathcal{F}_{\text{I}^t}^{\text{aut}}$. In fact, this should correspond to the second assertion of Theorem 6.3.1. However, the general duality compatibility between $\mathbb{D}_{\text{Verd}}$ and $\mathbb{D}_{\text{tw.adm}}$, as conjectured in [Han23, Conjecture 3.1.4], still remains open. Note that for general G , this duality compatibility states that along the restricted (conjectural) equivalence $\mathbf{L}_\psi: \text{D}(\text{Bun}_G)^{\text{ULA}} \xrightarrow{\sim} \text{Adm}(\text{Par}_G)$, we have $\mathbb{D}_{\text{tw.adm}} \circ \mathbf{L}_\psi = \mathbf{L}_{\psi^{-1}} \circ \mathbb{D}_{\text{Verd}}$, for each Whittaker datum ψ . In our setting with $G = \text{GL}_n$, we choose ψ to be the standard Whittaker datum arising from generic character on the unipotent radical, so up to $G(E)$ -conjugacy, ψ and ψ^{-1} are identified.

Corollary 6.3.2. *Assume the duality compatibility for $G = \text{GL}_n$, that is, $\mathbb{D}_{\text{tw.adm}} \circ \mathbf{L}_G = \mathbf{L}_G \circ \mathbb{D}_{\text{Verd}}$ where $\mathbf{L}_G: \text{D}(\text{Bun}_G)^{\text{ULA}} \xrightarrow{\sim} \text{Adm}(\text{Par}_G)$. Then*

$$\mathbb{D}_{\text{Verd}} \mathcal{F}_{\text{I}}^{\text{aut}} = \mathcal{F}_{\text{I}^t}^{\text{aut}}.$$

Proof. Note that $\mathcal{F}_I^{\text{aut}}$ is ULA by Proposition 6.1.2 (2), with $\mathbf{L}_G(\mathcal{F}_I^{\text{aut}}) = \mathcal{F}_I \in \text{Adm}(\text{Par}_G)$ under the assumption. Then applying Theorem 6.3.1 immediately gives the claimed result. $\square$

REFERENCES

- [BM23] Alexander Bertoloni Meli. Coherent sheaves for the Steinberg parameter of PGL_2 , 2023. Unpublished notes. Available at <https://math.bu.edu/people/abertolo/PGL2notes.pdf>.
- [Dri74] Vladimir G. Drinfeld. Elliptic modules. *Mathematics of the USSR-Sbornik*, 23(4):561–592, 1974. English translation. doi:10.1070/SM1974v023n04ABEH001731.
- [Han23] David Hansen. Beijing notes on the categorical local Langlands conjecture, with an appendix by Adeel Khan, 2023. Last updated on May 2025; available at <http://www.davidrenshawhansen.net/Beijing.pdf>.
- [Hel23] Eugen Hellmann. On the derived category of the Iwahori–Hecke algebra. *Compositio Mathematica*, 159(5):1042–1110, 2023. doi:10.1112/S0010437X23007145.
- [HHS24] Linus Hamann, David Hansen, and Peter Scholze. Geometric Eisenstein series I: Finiteness theorems, 2024. Preprint. arXiv:2409.07363.
- [HKW22] David Hansen, Tasho Kaletha, and Jared Weinstein. On the Kottwitz conjecture for local shtuka spaces. *Forum of Mathematics, Pi*, 10:e13, 2022. doi:10.1017/fmp.2022.7.
- [HM26] David Hansen and Lucas Mann. Foundations for categorical local Langlands. In preparation, 2026.
- [RZ96] Michael Rapoport and Thomas Zink. *Period Spaces for p -divisible Groups*, volume 141 of *Annals of Mathematics Studies*. Princeton University Press, Princeton, NJ, 1996. doi:10.1515/9781400882601.
- [SS91] Peter Schneider and Ulrich Stuhler. The cohomology of p -adic symmetric spaces. *Inventiones Mathematicae*, 105(1):47–122, 1991. doi:10.1007/BF01232257.
- [Sta18] The Stacks Project Authors. *Stacks Project*. <https://stacks.math.columbia.edu>, 2018.
- [Yan25] Xiangqian Yang. On Ihara’s lemma for definite unitary groups, 2025. Preprint. arXiv:2504.07504.
- [YZ25] Xiangqian Yang and Xinwen Zhu. On the generic part of the cohomology of Shimura varieties of abelian type, 2025. Preprint. arXiv:2505.04329.
- [Zel80] Andrei V. Zelevinsky. Induced representations of reductive p -adic groups II: On irreducible representations of $\text{GL}(n)$. *Annales scientifiques de l’École Normale Supérieure*, 13(2):165–210, 1980. doi:10.24033/asens.1379.
- [Zhu25] Xinwen Zhu. Tame categorical local Langlands correspondence, 2025. Preprint. arXiv:2504.07482.